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(54) **ELECTRONIC DEVICE FOR MANAGING ACCESS TO AN APPLICATION AND METHOD THEREOF**

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CPC ***G06F 21/6218*** (2013.01)

(57) **ABSTRACT**

A method for managing access to content for an application of an electronic device includes receiving, from the application, a content access request for granting access to first data of one or more content types available via the electronic device; displaying one or more labels, from among a plurality of labels, corresponding to the one or more content types; receiving a first user selection of at least one label, from among the one or more labels, corresponding to the one or more content types; and granting, to the application, access to the first data of the one or more content types based on the first user selection.

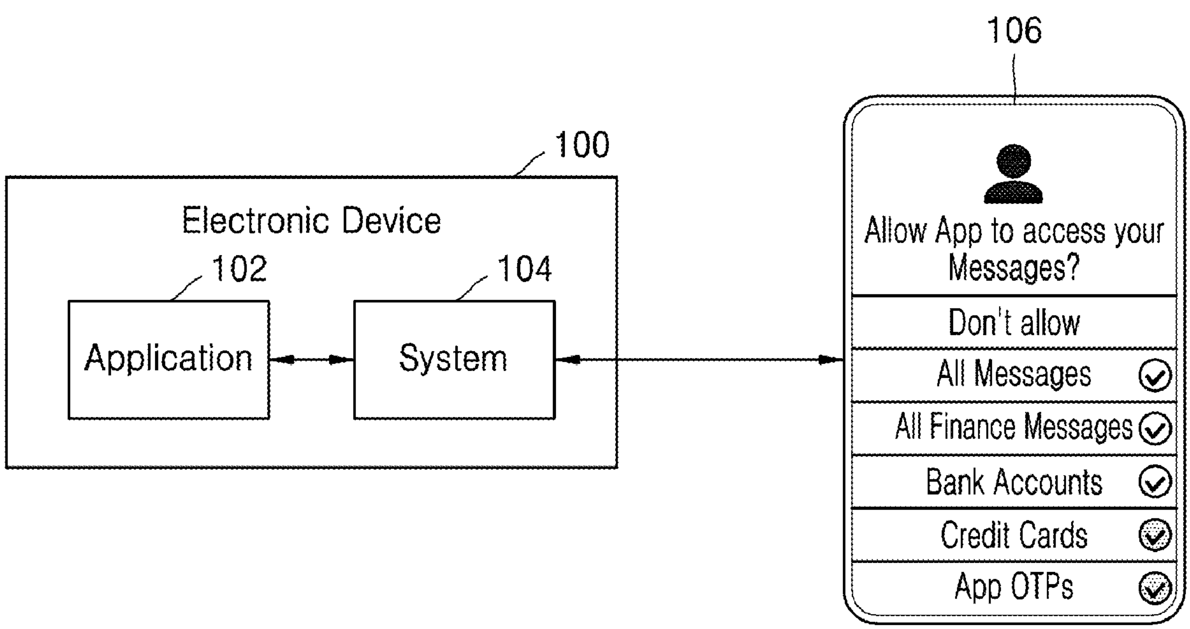


FIG. 1

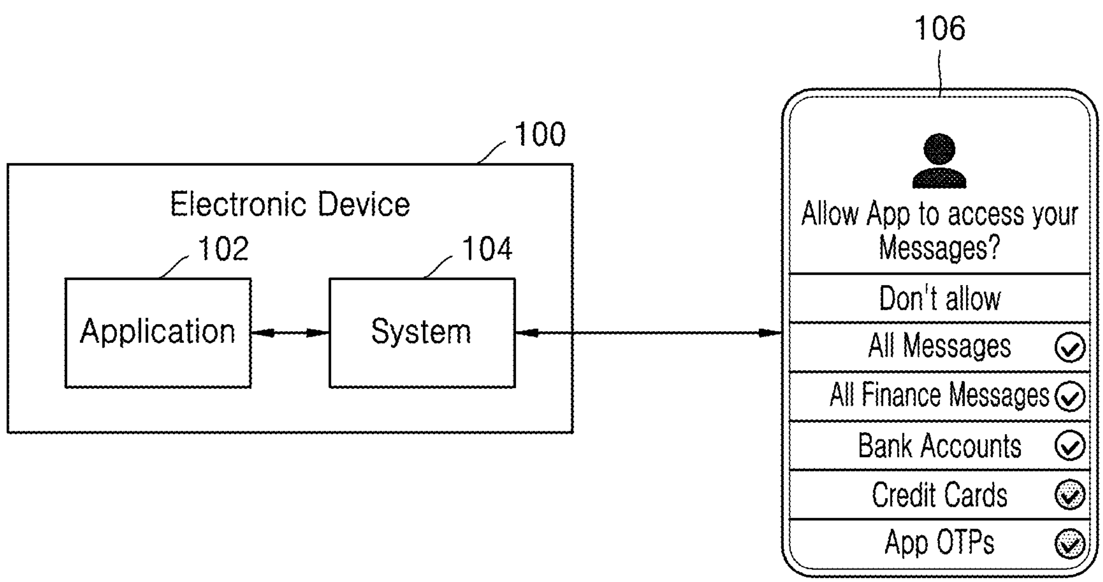

100
Electronic Device
102
Application
104
System
106
Allow App to access your Messages?
Don't allow
All Messages
All Finance Messages
Bank Accounts
Credit Cards
App OTPs

FIG. 2

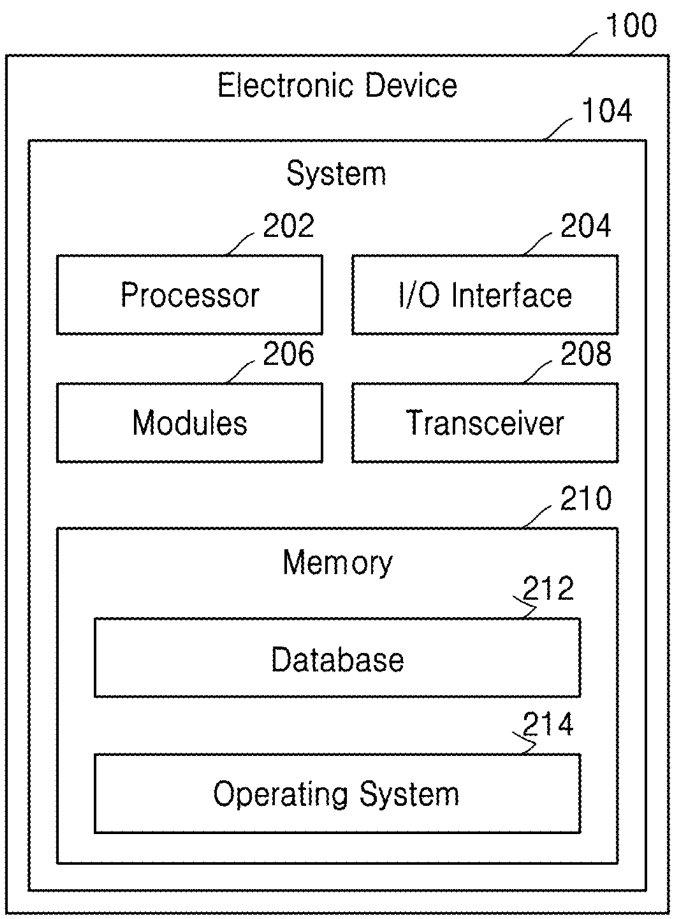

100
Electronic Device
104
System
202
Processor
204
I/O Interface
206
Modules
208
Transceiver
210
Memory
212
Database
214
Operating System

FIG. 3

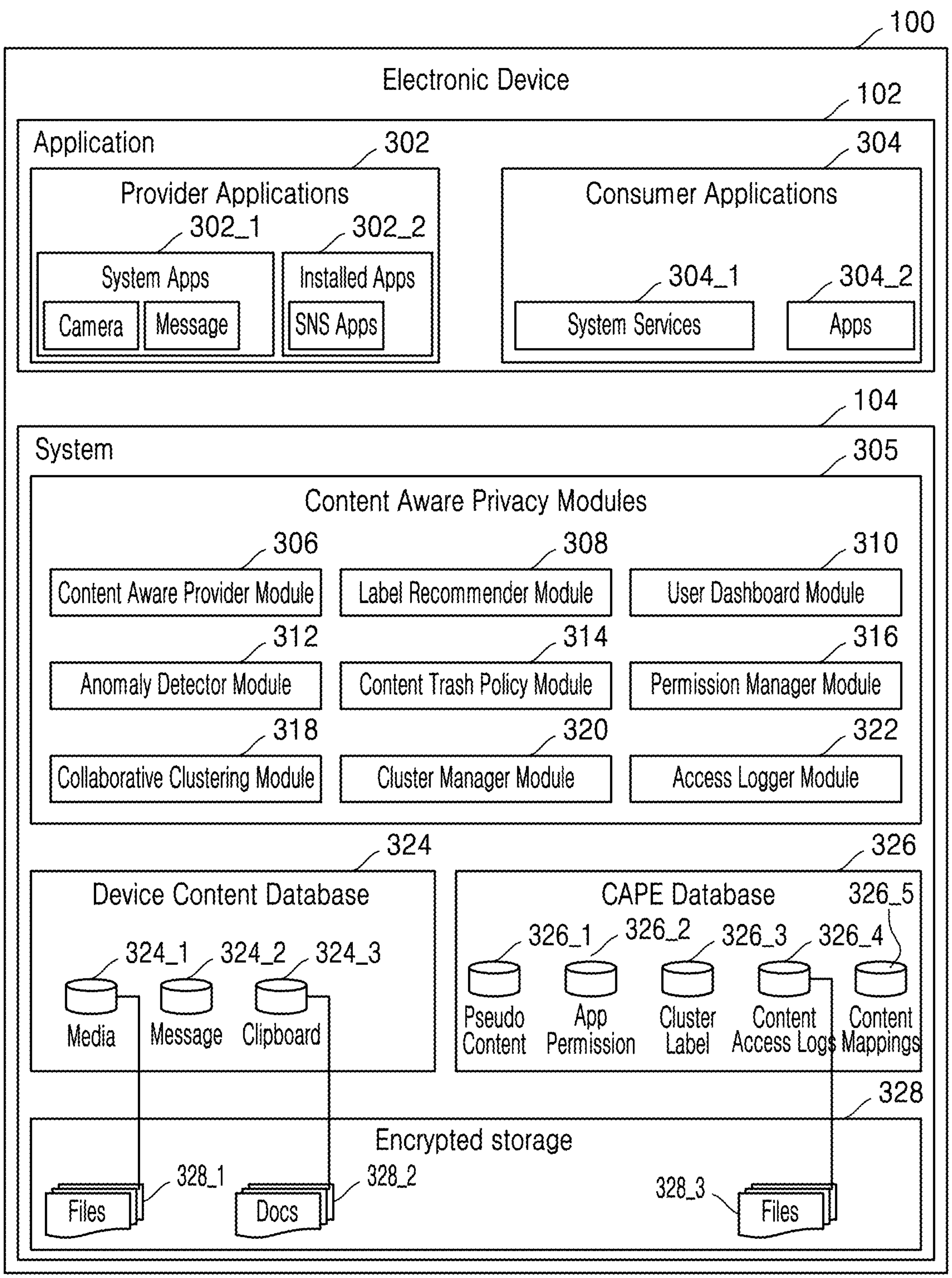

100
Electronic Device
102
Application
302
Provider Applications
302_1
302_2
System Apps
Installed Apps
Camera
Message
SNS Apps
304
Consumer Applications
304_1
304_2
System Services
Apps
104
System
305
Content Aware Privacy Modules
306
Content Aware Provider Module
308
Label Recommender Module
310
User Dashboard Module
312
Anomaly Detector Module
314
Content Trash Policy Module
316
Permission Manager Module
318
Collaborative Clustering Module
320
Cluster Manager Module
322
Access Logger Module
324
Device Content Database
324_1
324_2
324_3
Media
Message
Clipboard
326
CAPE Database
326_1
326_2
326_3
326_4
326_5
Pseudo Content
App Permission
Cluster Label
Content Access Logs
Content Mappings
328
Encrypted storage
328_1
Files
328_2
Docs
328_3
Files

FIG. 4

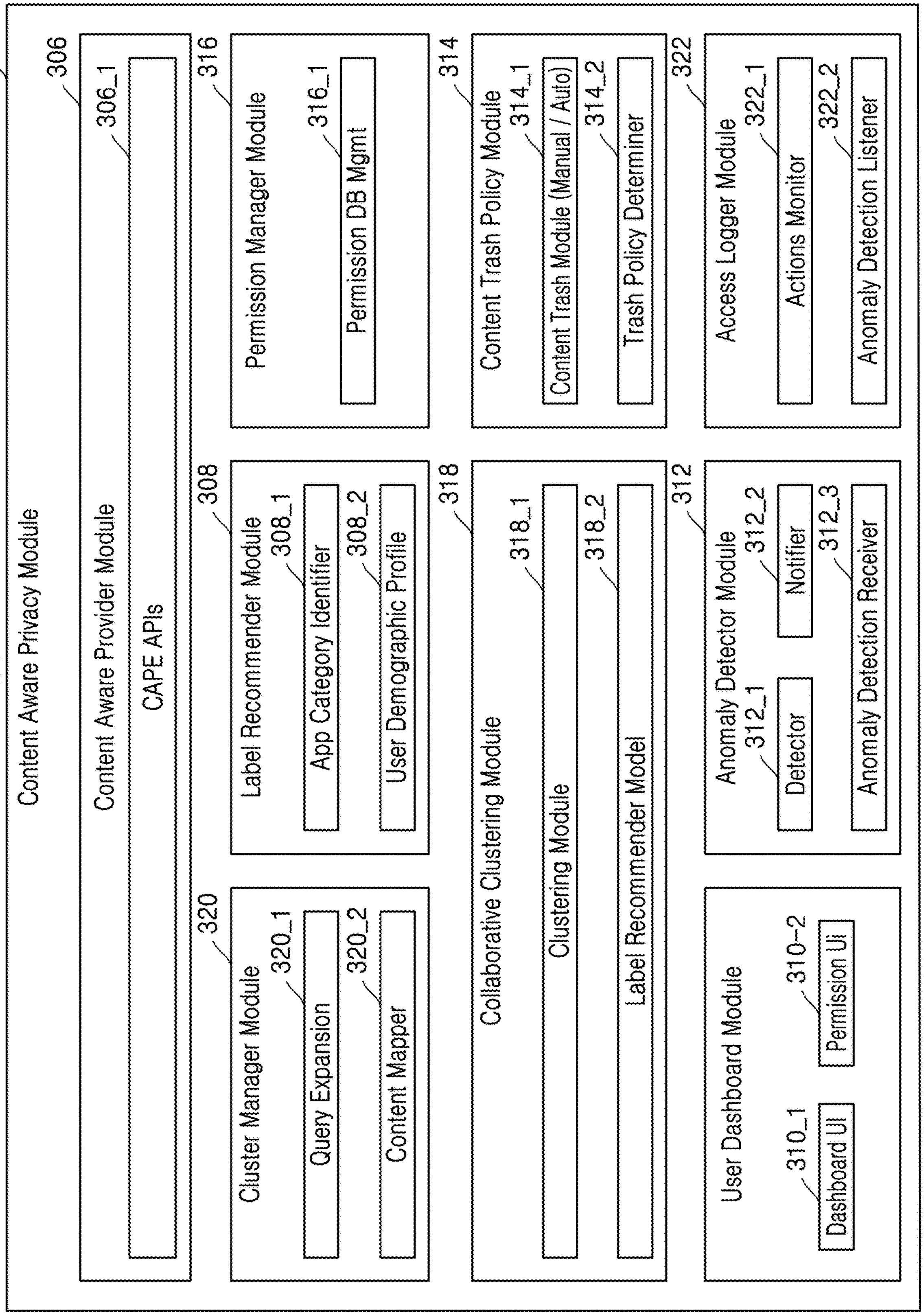

305
Content Aware Privacy Module
306
Content Aware Provider Module
306_1
CAPE APIs
320
Cluster Manager Module
320_1
Query Expansion
320_2
Content Mapper
308
Label Recommender Module
308_1
App Category Identifier
308_2
User Demographic Profile
316
Permission Manager Module
316_1
Permission DB Mgmt
318
Collaborative Clustering Module
318_1
Clustering Module
318_2
Label Recommender Model
314
Content Trash Policy Module
314_1
Content Trash Module (Manual / Auto)
314_2
Trash Policy Determiner
User Dashboard Module
310_1
Dashboard UI
310-2
Permission Ui
312
Anomaly Detector Module
312_1
Detector
312_2
Notifier
312_3
Anomaly Detection Receiver
322
Access Logger Module
322_1
Actions Monitor
322_2
Anomaly Detection Listener

FIG. 5

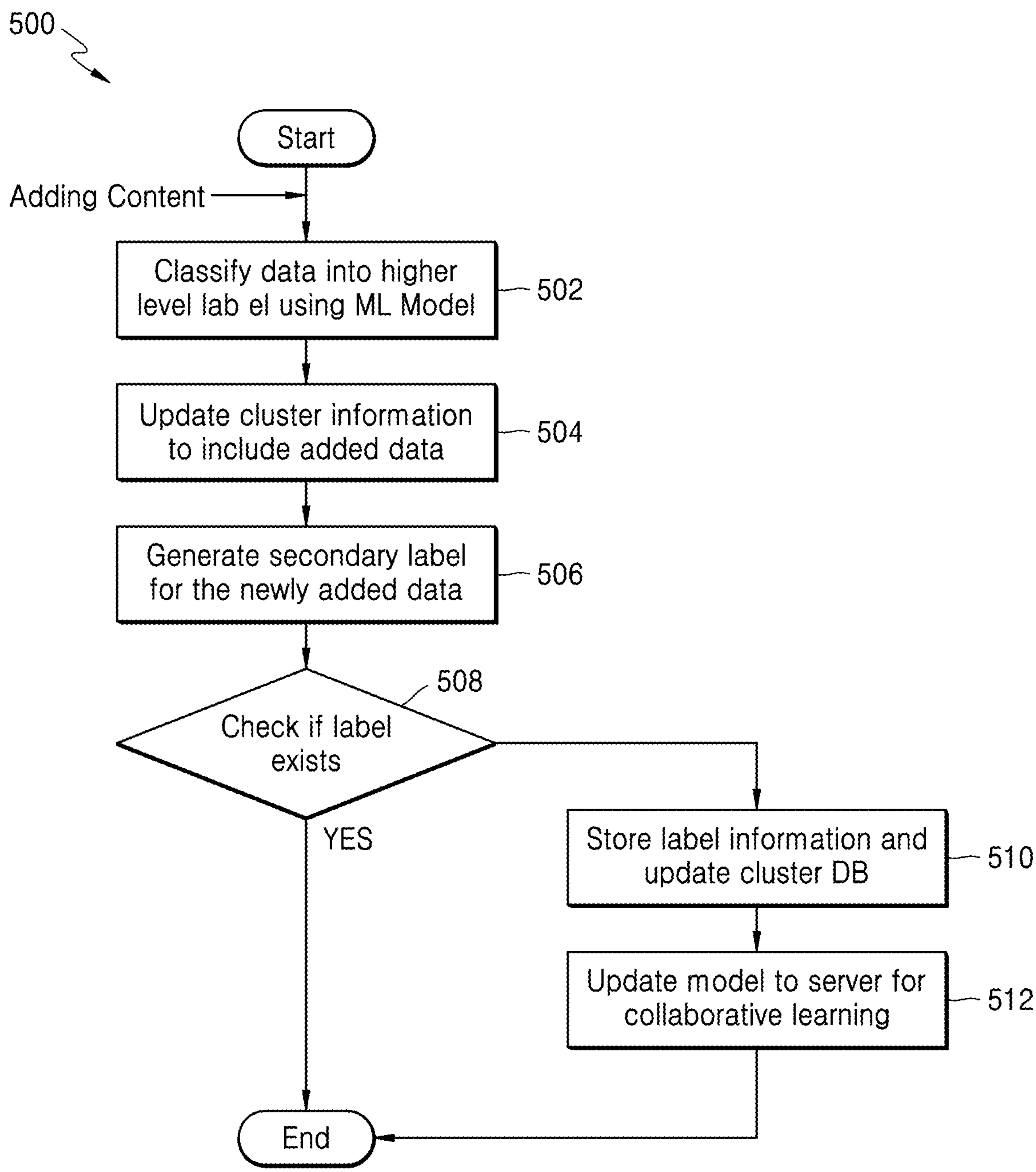

500
Start
Adding Content
Classify data into higher level lab el using ML Model
502
Update cluster information to include added data
504
Generate secondary label for the newly added data
506
Check if label exists
508
YES
Store label information and update cluster DB
510
Update model to server for collaborative learning
512
End

FIG. 6

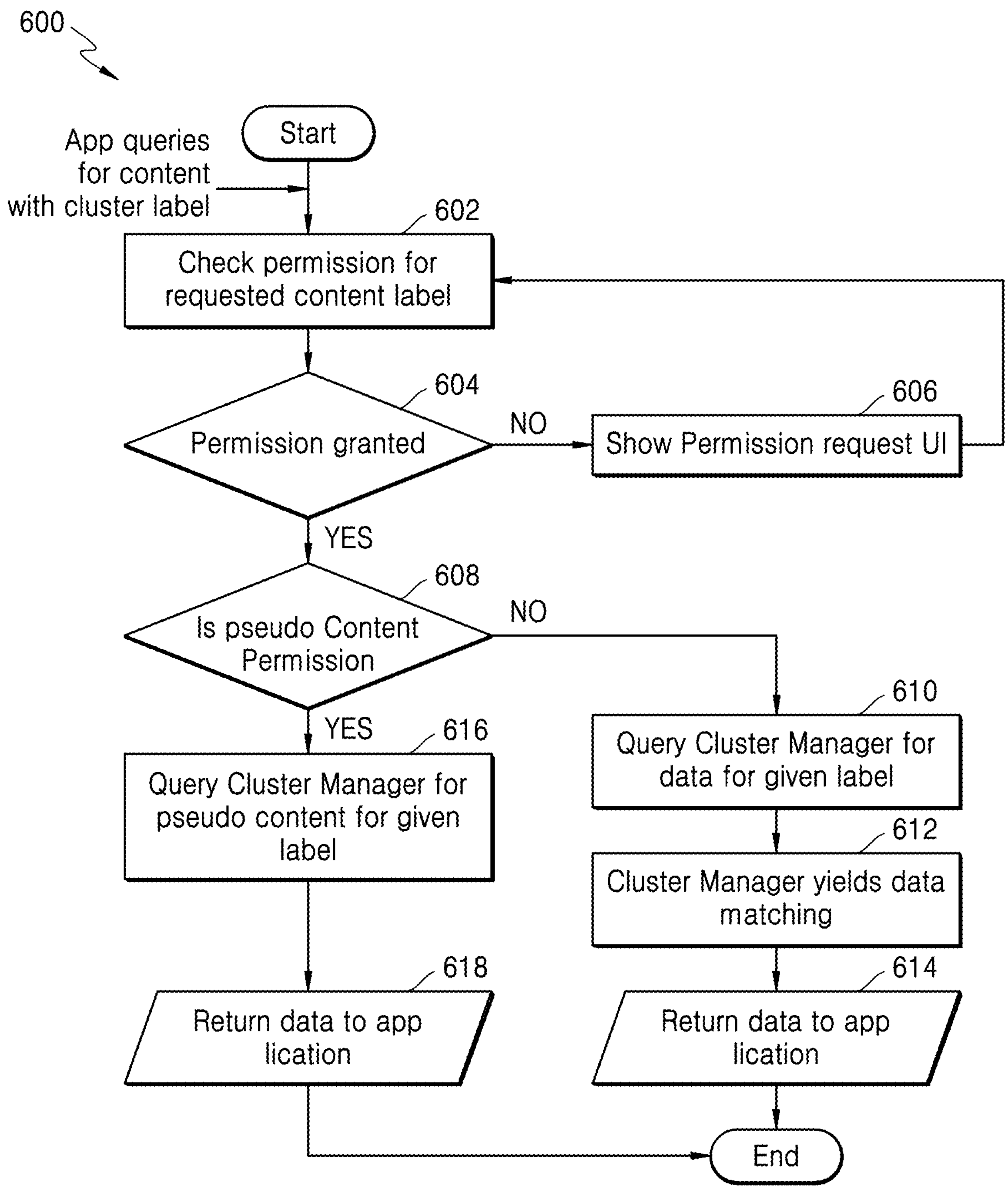

600
Start
App queries
for content
with cluster label
602
Check permission for
requested content label
604
Permission granted
NO
606
Show Permission request UI
YES
608
Is pseudo Content
Permission
NO
610
Query Cluster Manager for
data for given label
YES
616
Query Cluster Manager for
pseudo content for given
label
612
Cluster Manager yields data
matching
618
Return data to app
lication
614
Return data to app
lication
End

FIG. 7

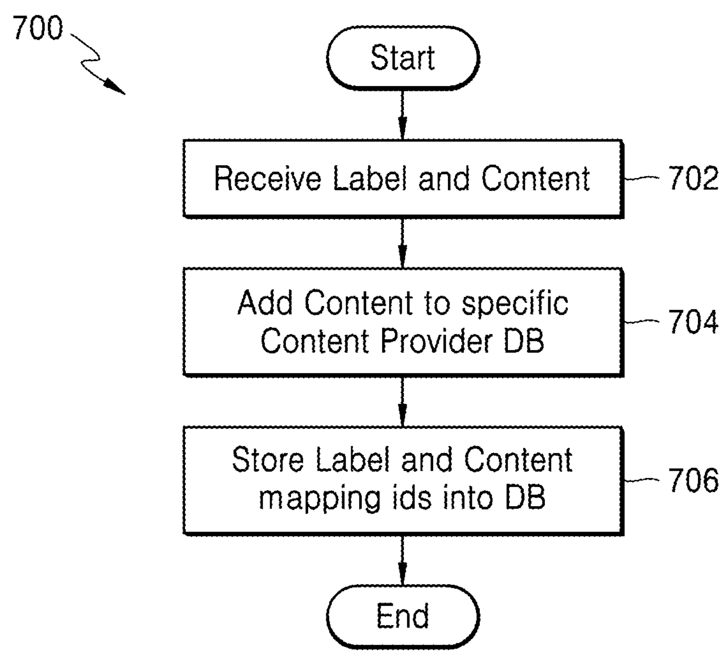

700
Start
Receive Label and Content
702
Add Content to specific Content Provider DB
704
Store Label and Content mapping ids into DB
706
End

FIG. 8

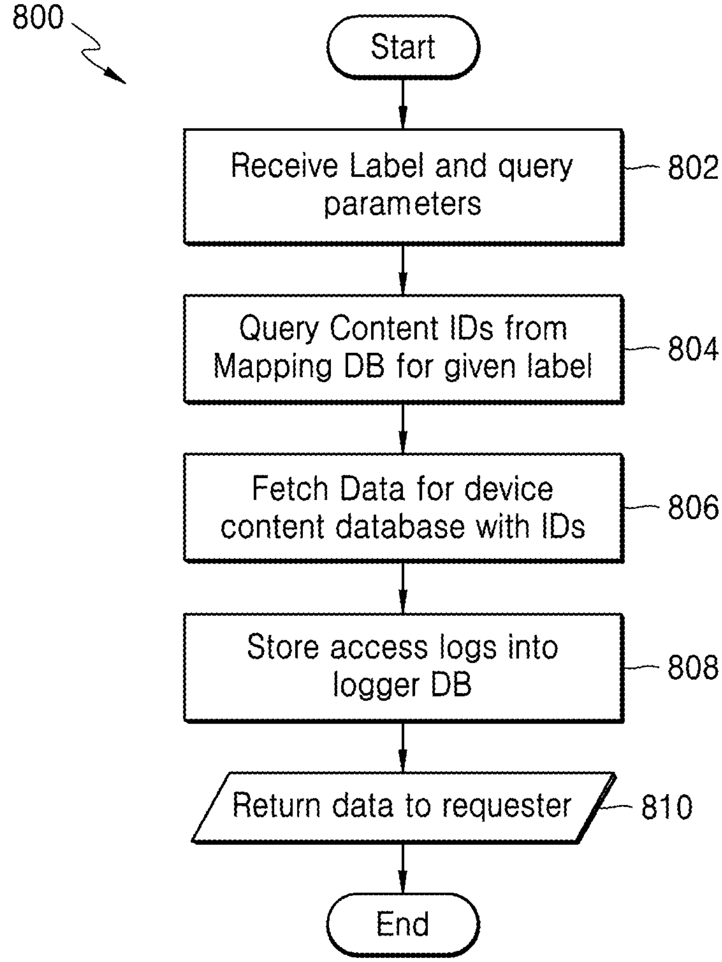

800
Start
Receive Label and query parameters
802
Query Content IDs from Mapping DB for given label
804
Fetch Data for device content database with IDs
806
Store access logs into logger DB
808
Return data to requester
810
End

FIG. 9

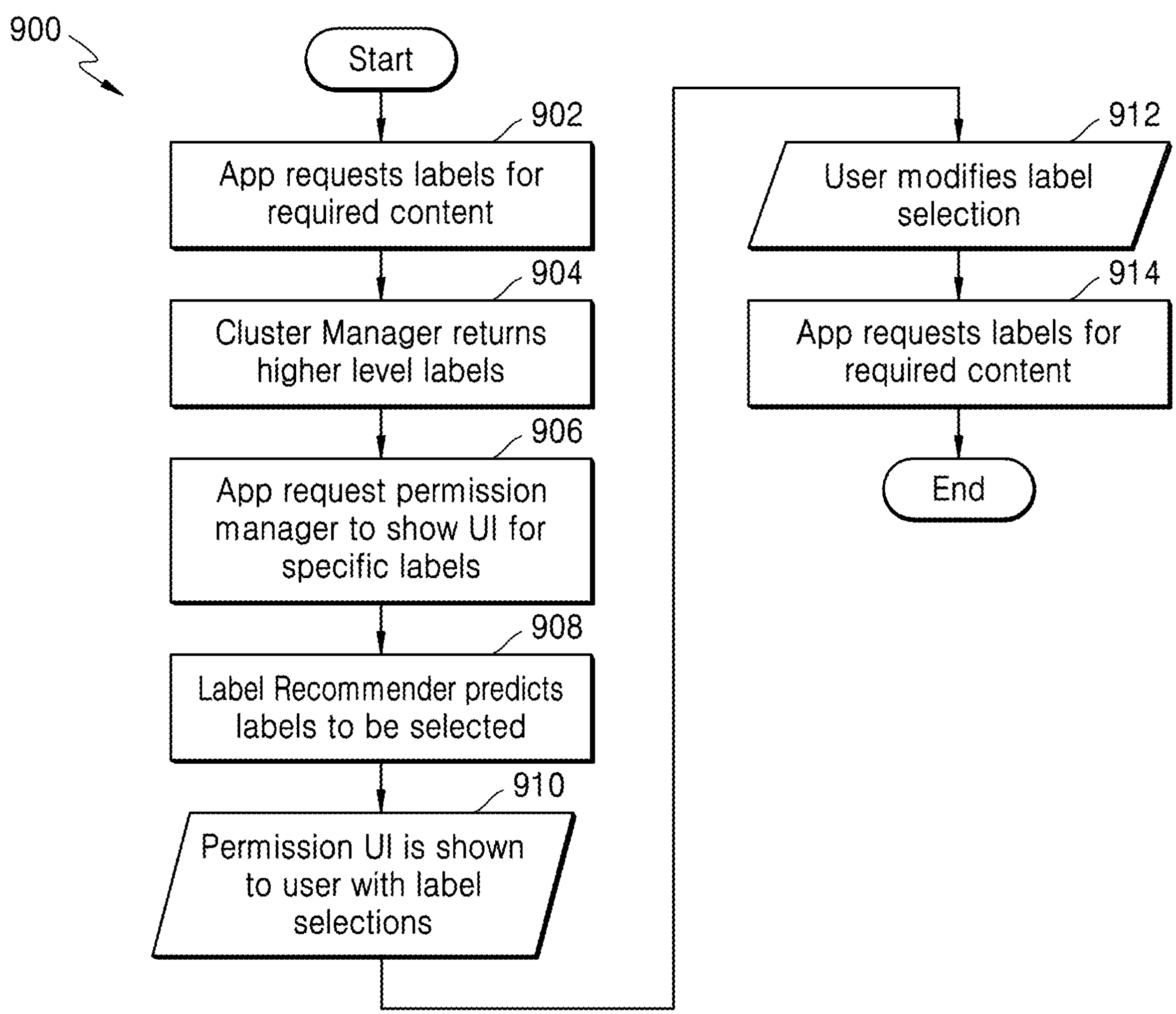

900
Start
902
App requests labels for required content
904
Cluster Manager returns higher level labels
906
App request permission manager to show UI for specific labels
908
Label Recommender predicts labels to be selected
910
Permission UI is shown to user with label selections
912
User modifies label selection
914
App requests labels for required content
End

FIG. 10

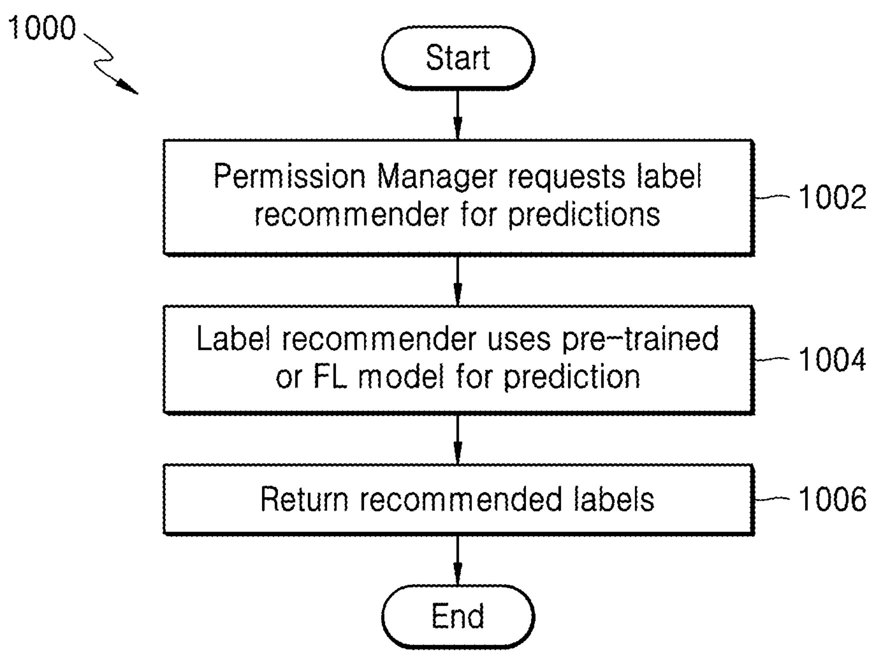

1000
Start
Permission Manager requests label recommender for predictions
1002
Label recommender uses pre-trained or FL model for prediction
1004
Return recommended labels
1006
End

FIG. 11

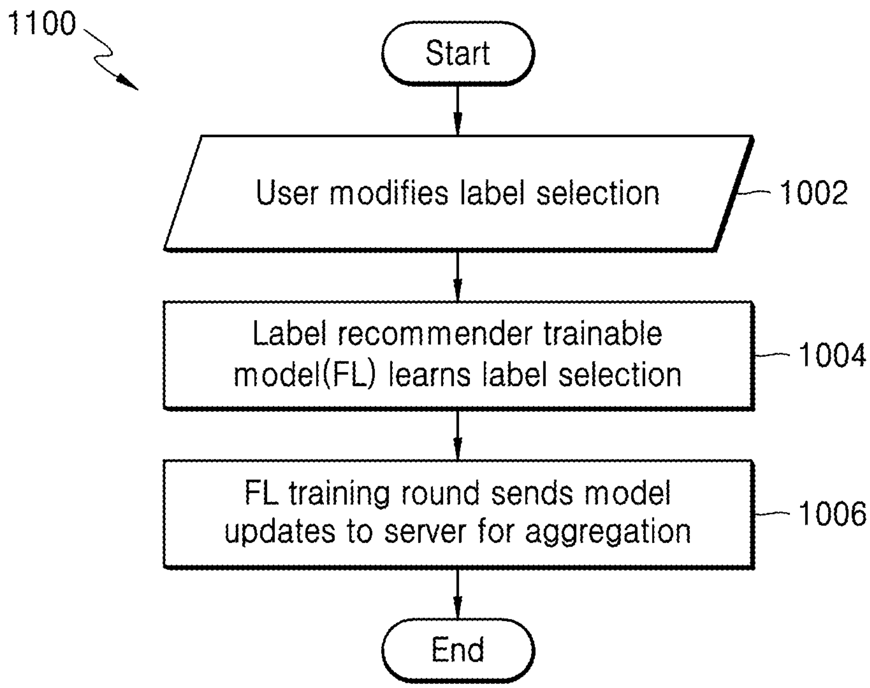

1100
Start
User modifies label selection
1002
Label recommender trainable model(FL) learns label selection
1004
FL training round sends model updates to server for aggregation
1006
End

FIG. 12

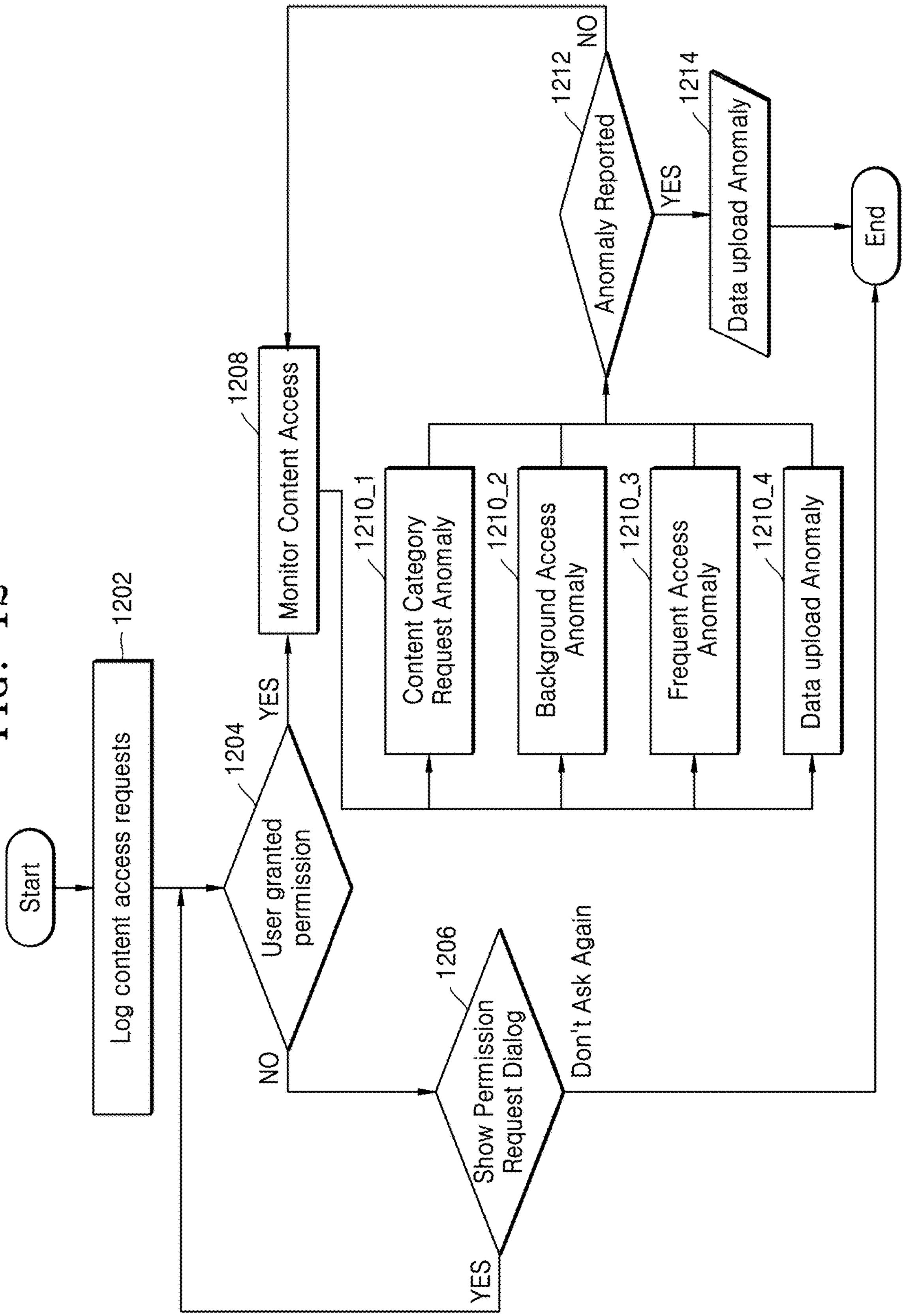

Start
Log content access requests
1202
User granted permission
1204
YES
NO
Show Permission Request Dialog
1206
YES
Don't Ask Again
Monitor Content Access
1208
Content Category Request Anomaly
1210_1
Background Access Anomaly
1210_2
Frequent Access Anomaly
1210_3
Data upload Anomaly
1210_4
Anomaly Reported
1212
YES
NO
Data upload Anomaly
1214
End

FIG. 13

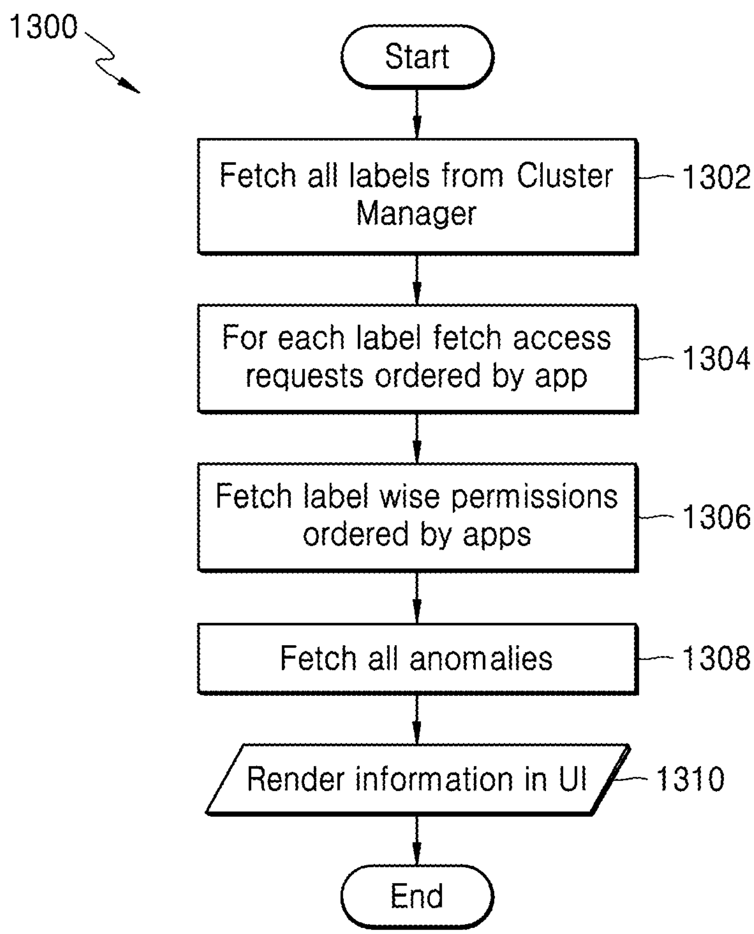

1300
Start
Fetch all labels from Cluster Manager
1302
For each label fetch access requests ordered by app
1304
Fetch label wise permissions ordered by apps
1306
Fetch all anomalies
1308
Render information in UI
1310
End

FIG. 14

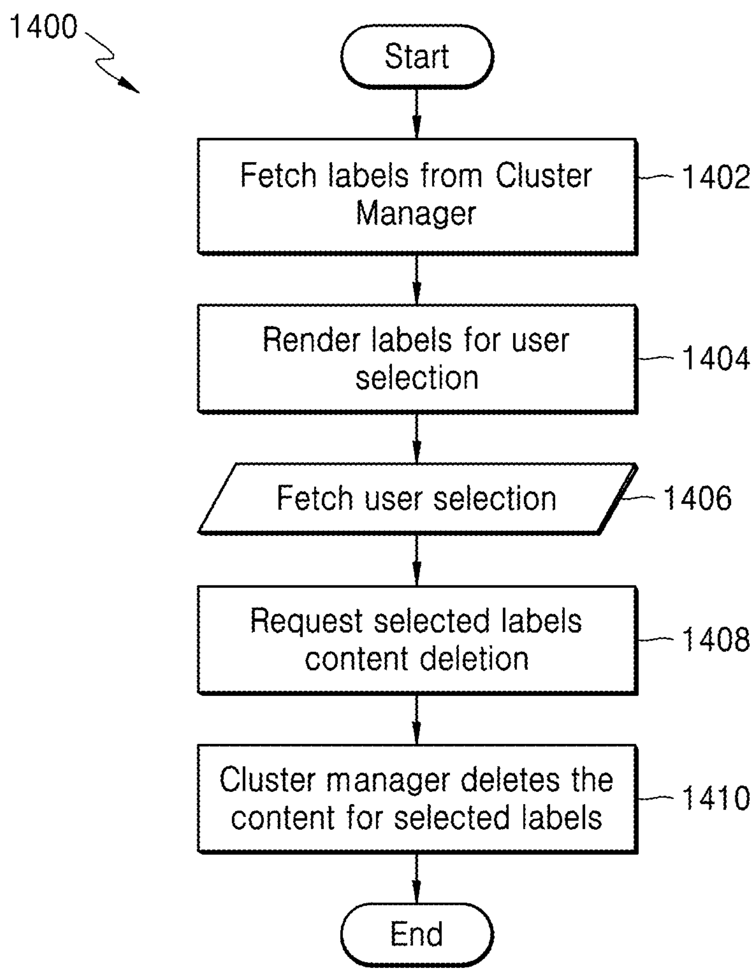

1400
Start
Fetch labels from Cluster Manager
1402
Render labels for user selection
1404
Fetch user selection
1406
Request selected labels content deletion
1408
Cluster manager deletes the content for selected labels
1410
End

FIG. 15A

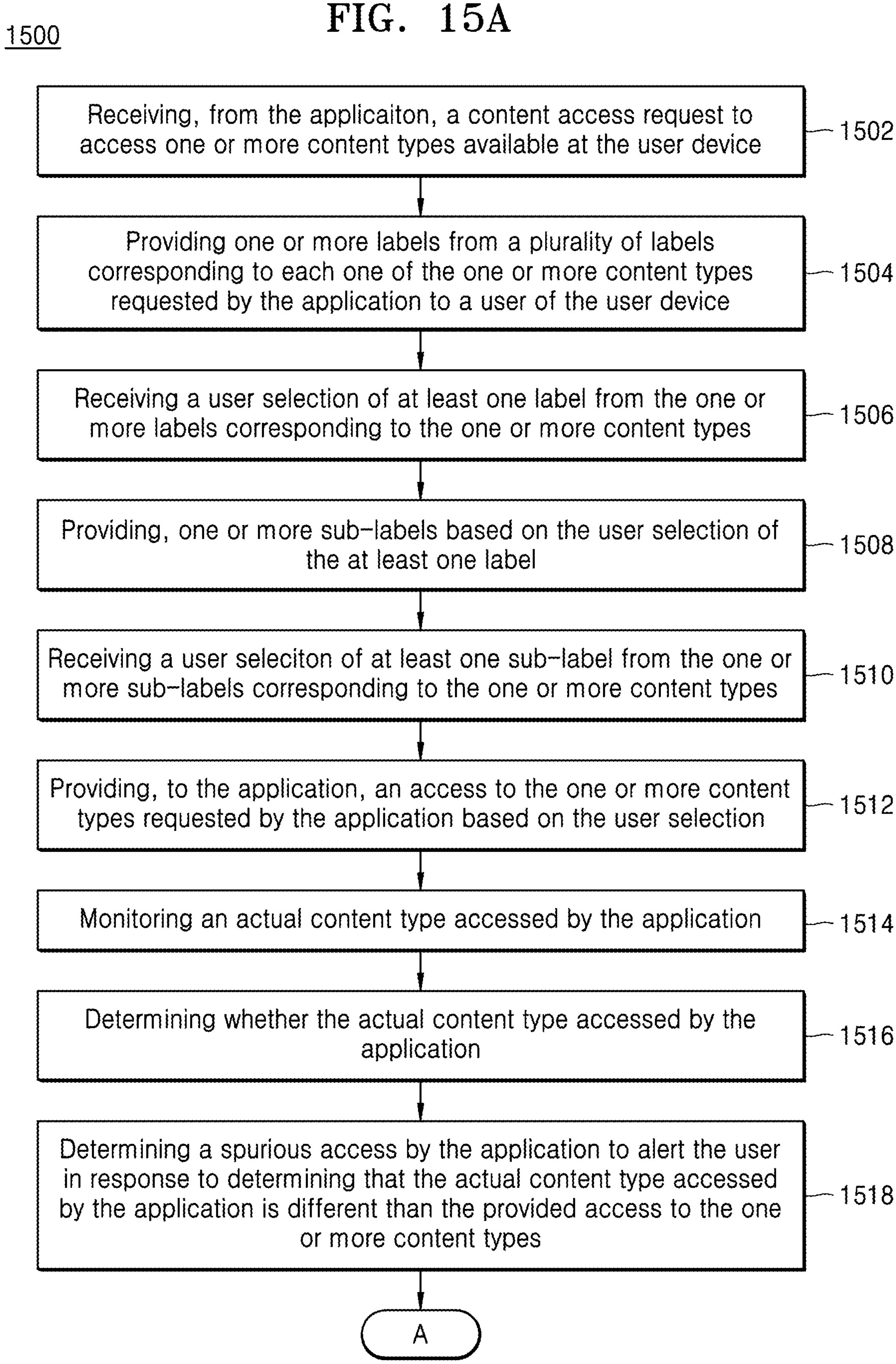

1500
Receiving, from the applicaiton, a content access request to access one or more content types available at the user device
1502
Providing one or more labels from a plurality of labels corresponding to each one of the one or more content types requested by the application to a user of the user device
1504
Receiving a user selection of at least one label from the one or more labels corresponding to the one or more content types
1506
Providing, one or more sub-labels based on the user selection of the at least one label
1508
Receiving a user seleciton of at least one sub-label from the one or more sub-labels corresponding to the one or more content types
1510
Providing, to the application, an access to the one or more content types requested by the application based on the user selection
1512
Monitoring an actual content type accessed by the application
1514
Determining whether the actual content type accessed by the application
1516
Determining a spurious access by the application to alert the user in response to determining that the actual content type accessed by the application is different than the provided access to the one or more content types
1518
A

FIG. 15B

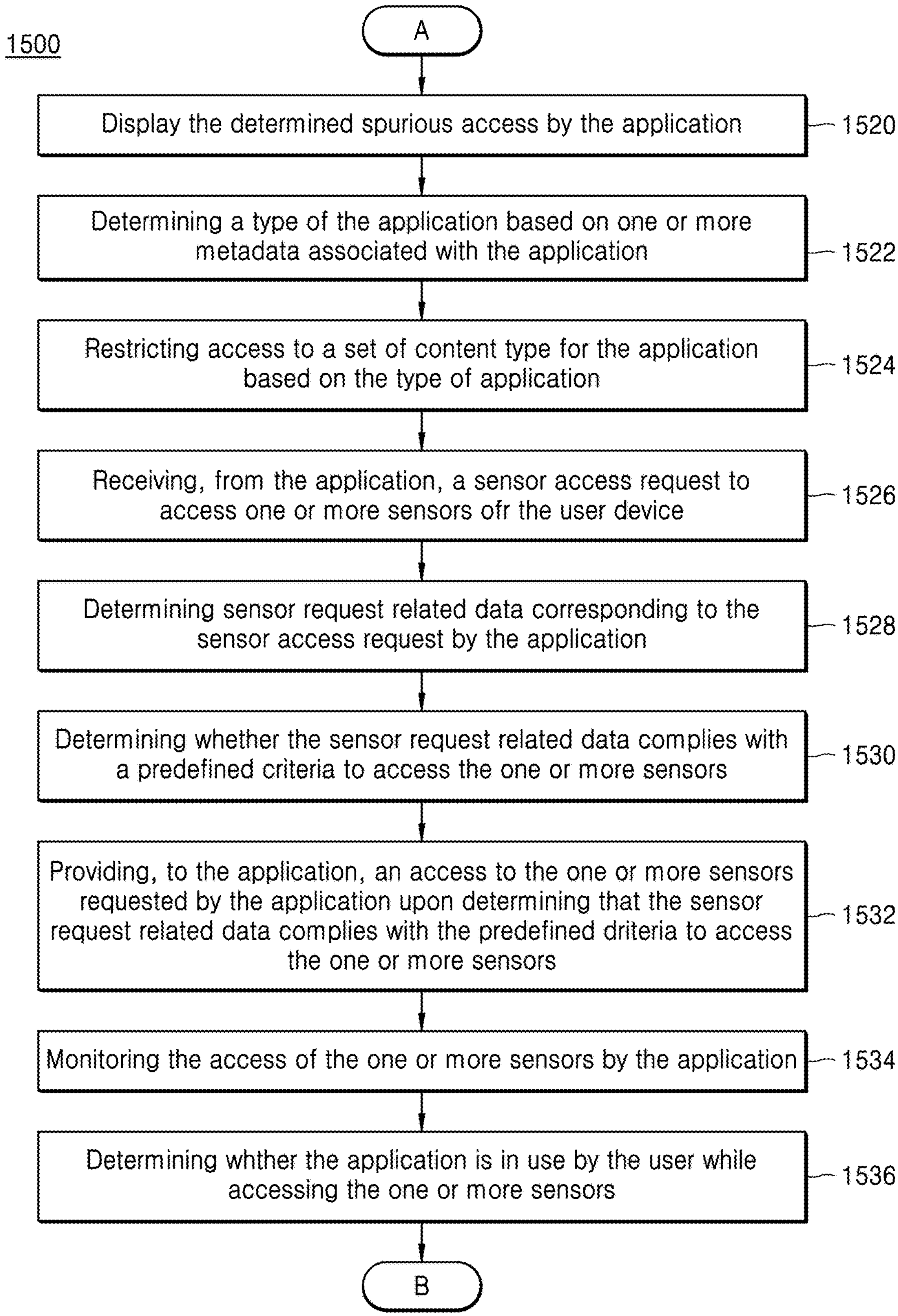

1500
A
Display the determined spurious access by the application
1520
Determining a type of the application based on one or more metadata associated with the application
1522
Restricting access to a set of content type for the application based on the type of application
1524
Receiving, from the application, a sensor access request to access one or more sensors ofr the user device
1526
Determining sensor request related data corresponding to the sensor access request by the application
1528
Determining whether the sensor request related data complies with a predefined criteria to access the one or more sensors
1530
Providing, to the application, an access to the one or more sensors requested by the application upon determining that the sensor request related data complies with the predefined driteria to access the one or more sensors
1532
Monitoring the access of the one or more sensors by the application
1534
Determining whther the application is in use by the user while accessing the one or more sensors
1536
B

FIG. 15C

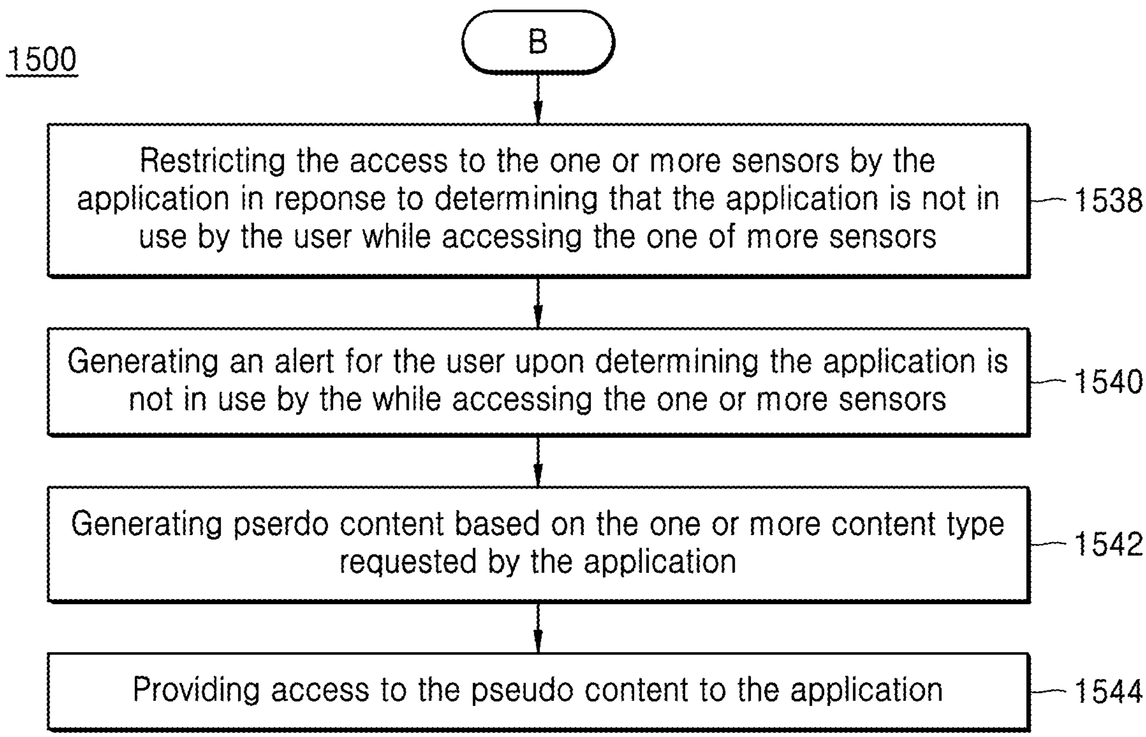

1500
B
Restricting the access to the one or more sensors by the application in reponse to determining that the application is not in use by the user while accessing the one of more sensors
1538
Generating an alert for the user upon determining the application is not in use by the while accessing the one or more sensors
1540
Generating pserdo content based on the one or more content type requested by the application
1542
Providing access to the pseudo content to the application
1544

FIG. 16A

1600

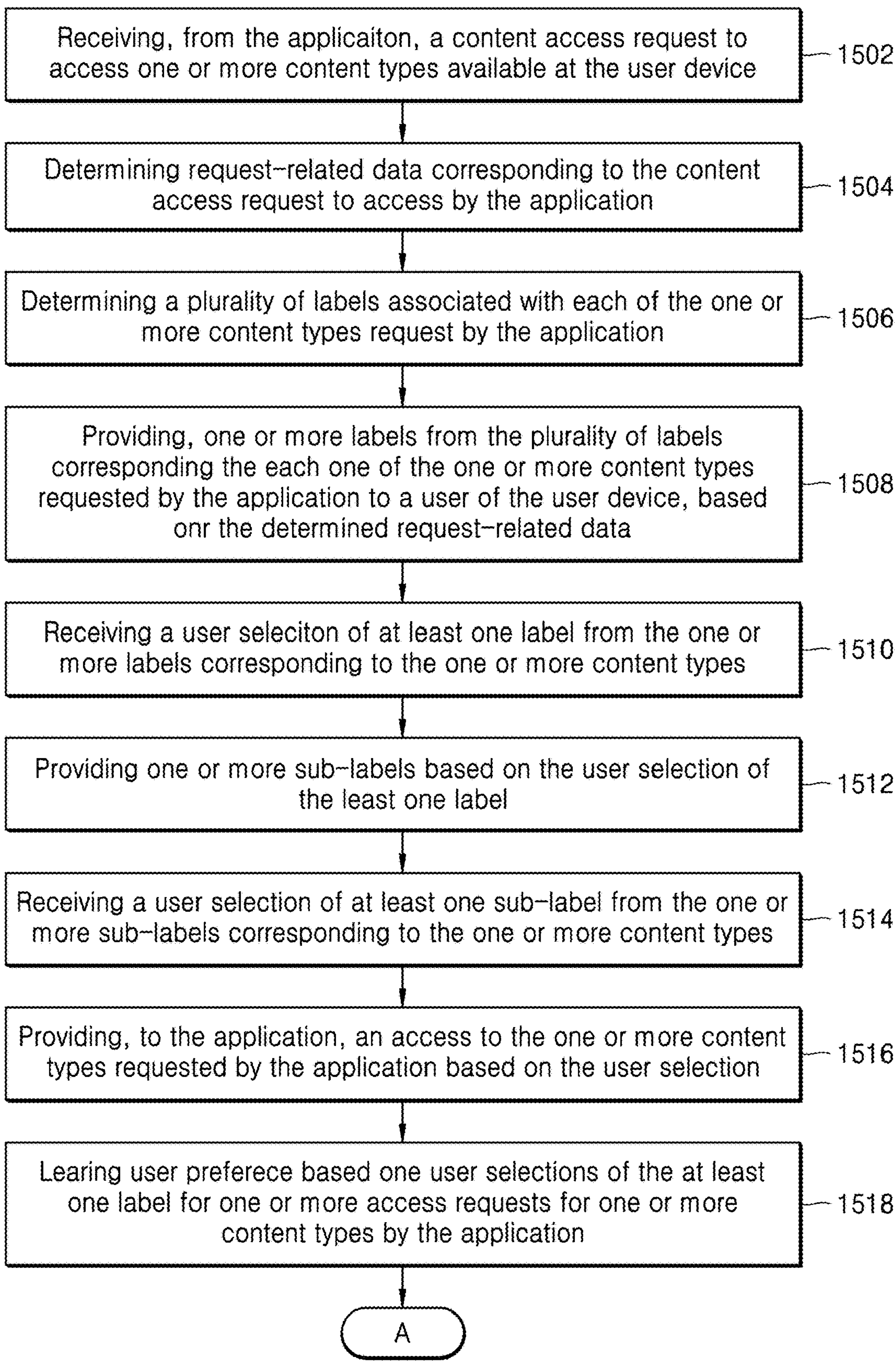

Receiving, from the applicaiton, a content access request to access one or more content types available at the user device
1502
Determining request-related data corresponding to the content access request to access by the application
1504
Determining a plurality of labels associated with each of the one or more content types request by the application
1506
Providing, one or more labels from the plurality of labels corresponding the each one of the one or more content types requested by the application to a user of the user device, based onr the determined request-related data
1508
Receiving a user seleciton of at least one label from the one or more labels corresponding to the one or more content types
1510
Providing one or more sub-labels based on the user selection of the least one label
1512
Receiving a user selection of at least one sub-label from the one or more sub-labels corresponding to the one or more content types
1514
Providing, to the application, an access to the one or more content types requested by the application based on the user selection
1516
Learing user preferece based one user selections of the at least one label for one or more access requests for one or more content types by the application
1518
A

FIG. 16B

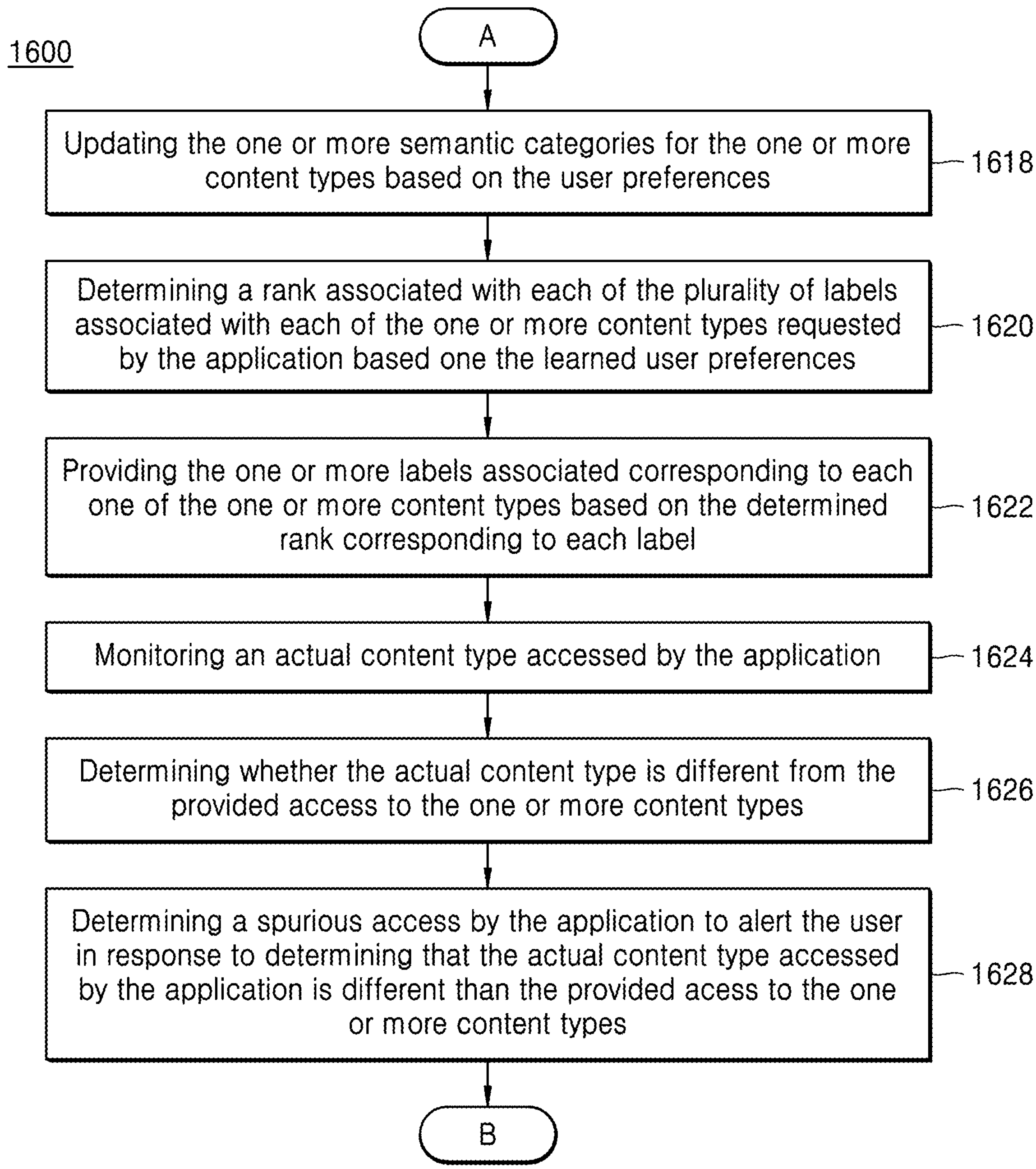

1600
A
Updating the one or more semantic categories for the one or more content types based on the user preferences
1618
Determining a rank associated with each of the plurality of labels associated with each of the one or more content types requested by the application based one the learned user preferences
1620
Providing the one or more labels associated corresponding to each one of the one or more content types based on the determined rank corresponding to each label
1622
Monitoring an actual content type accessed by the application
1624
Determining whether the actual content type is different from the provided access to the one or more content types
1626
Determining a spurious access by the application to alert the user in response to determining that the actual content type accessed by the application is different than the provided acess to the one or more content types
1628
B

FIG. 16C

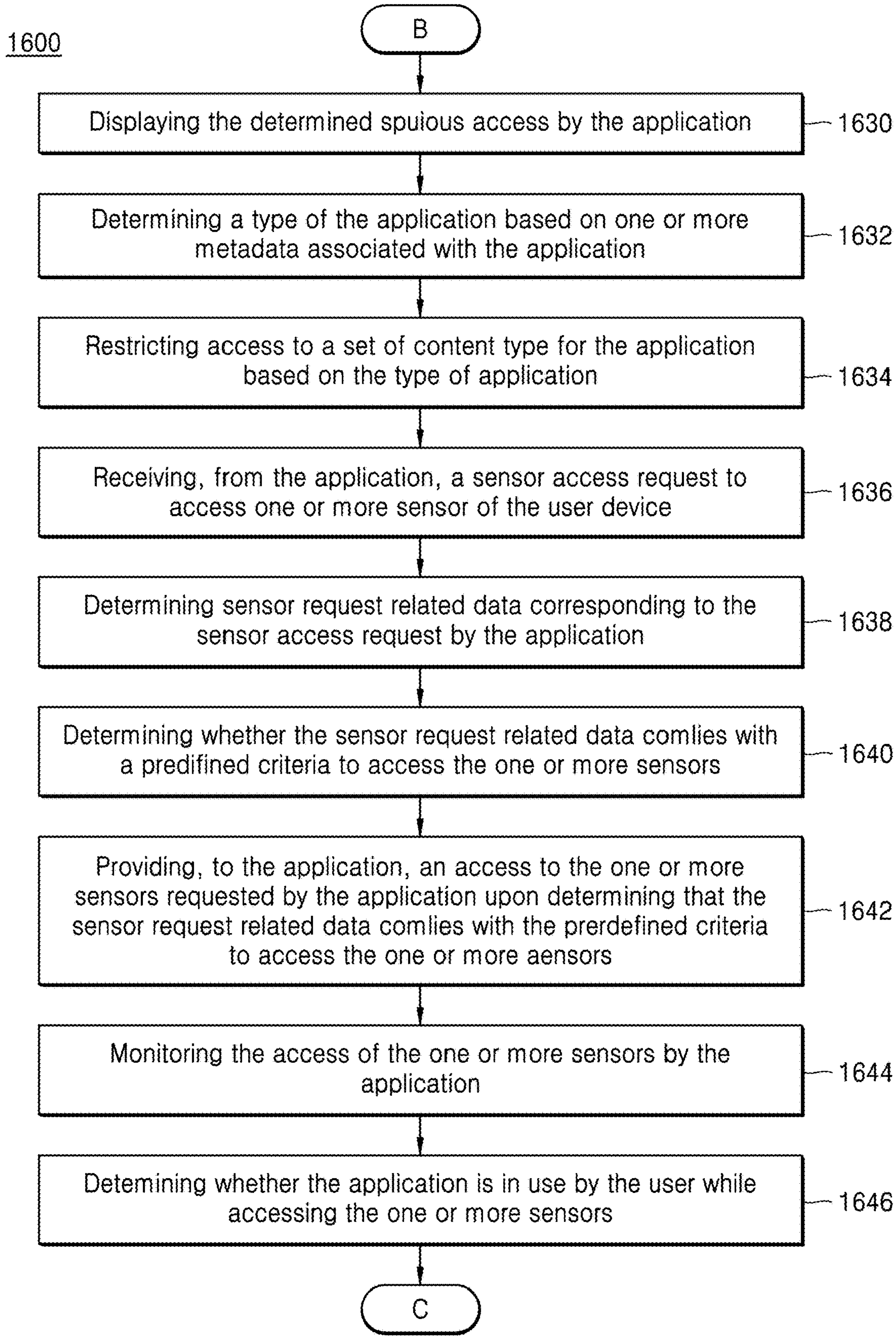

1600
B
Displaying the determined spuious access by the application
1630
Determining a type of the application based on one or more metadata associated with the application
1632
Restricting access to a set of content type for the application based on the type of application
1634
Receiving, from the application, a sensor access request to access one or more sensor of the user device
1636
Determining sensor request related data corresponding to the sensor access request by the application
1638
Determining whether the sensor request related data comlies with a predifined criteria to access the one or more sensors
1640
Providing, to the application, an access to the one or more sensors requested by the application upon determining that the sensor request related data comlies with the prerdefined criteria to access the one or more aensors
1642
Monitoring the access of the one or more sensors by the application
1644
Detemining whether the application is in use by the user while accessing the one or more sensors
1646
C

FIG. 16D

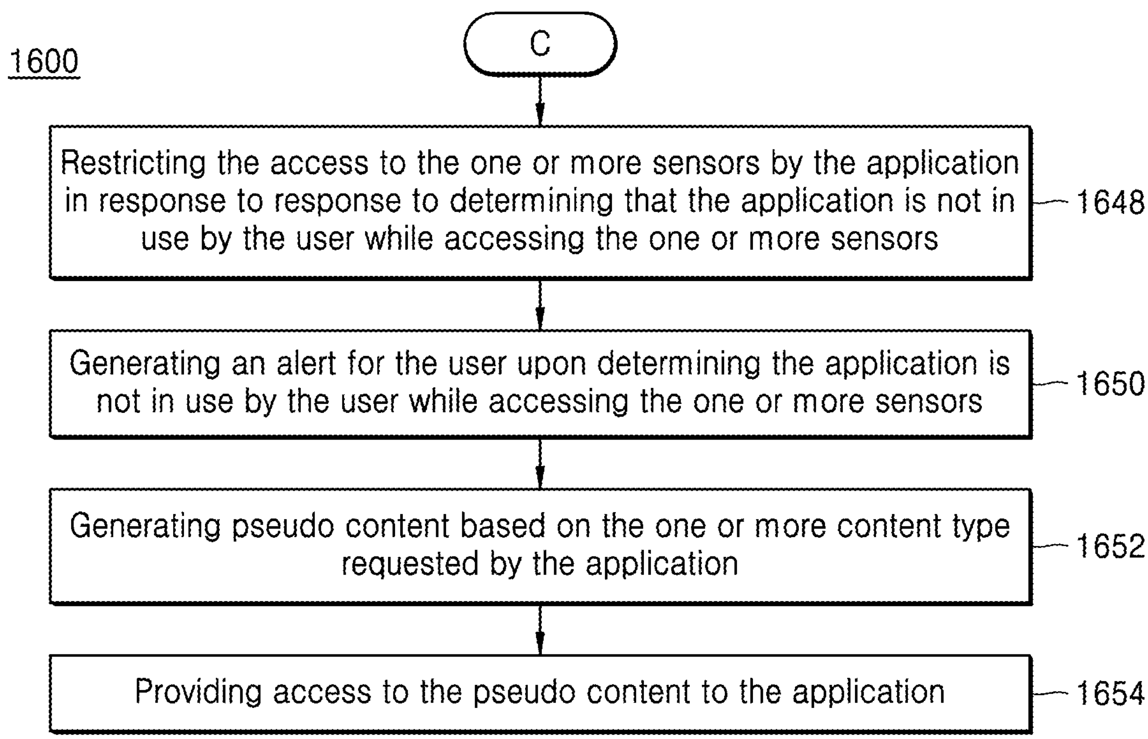

1600
C
Restricting the access to the one or more sensors by the application in response to response to determining that the application is not in use by the user while accessing the one or more sensors
1648
Generating an alert for the user upon determining the application is not in use by the user while accessing the one or more sensors
1650
Generating pseudo content based on the one or more content type requested by the application
1652
Providing access to the pseudo content to the application
1654

FIG. 17

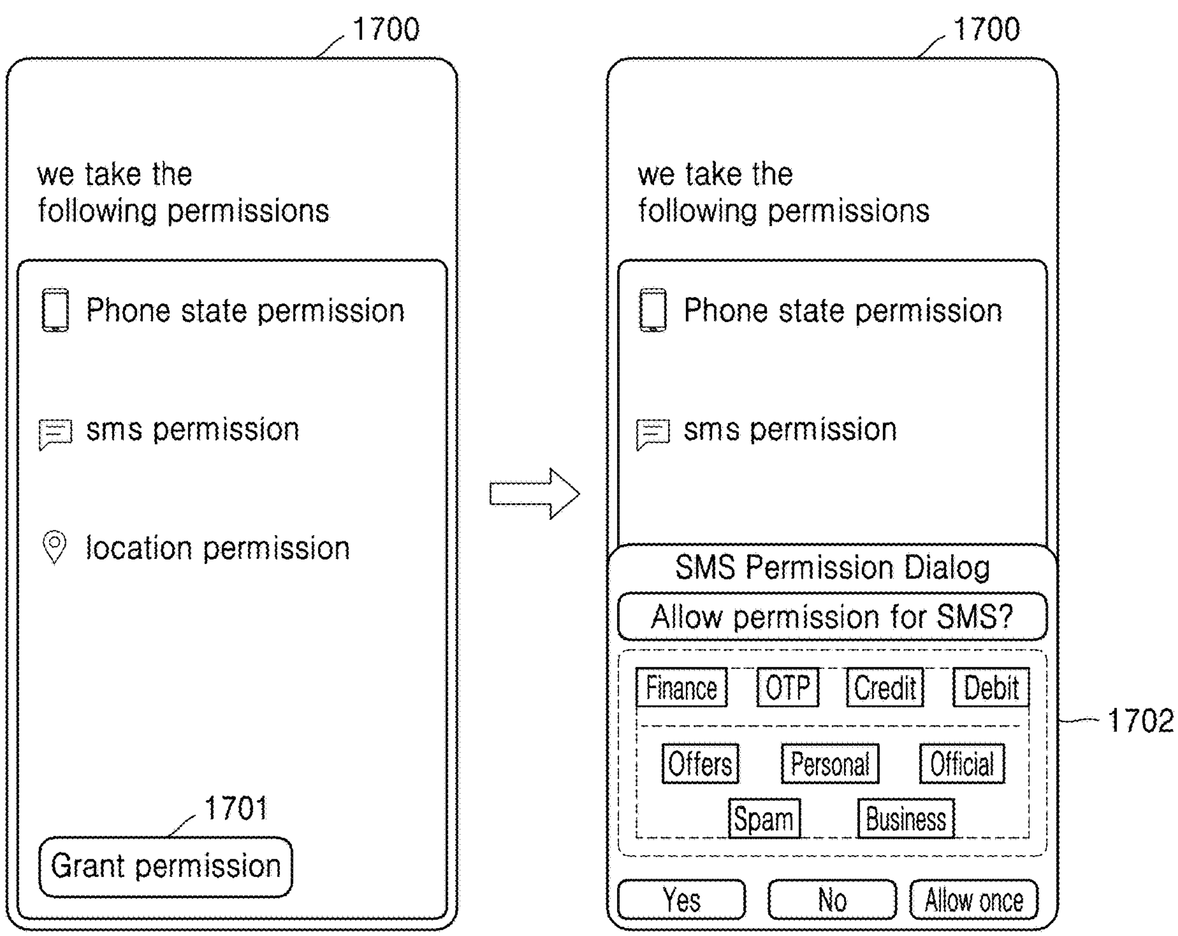

1700
we take the
following permissions
Phone state permission
sms permission
location permission
1701
Grant permission
1700
we take the
following permissions
Phone state permission
sms permission
SMS Permission Dialog
Allow permission for SMS?
Finance
OTP
Credit
Debit
Offers
Personal
Official
Spam
Business
1702
Yes
No
Allow once

FIG. 18

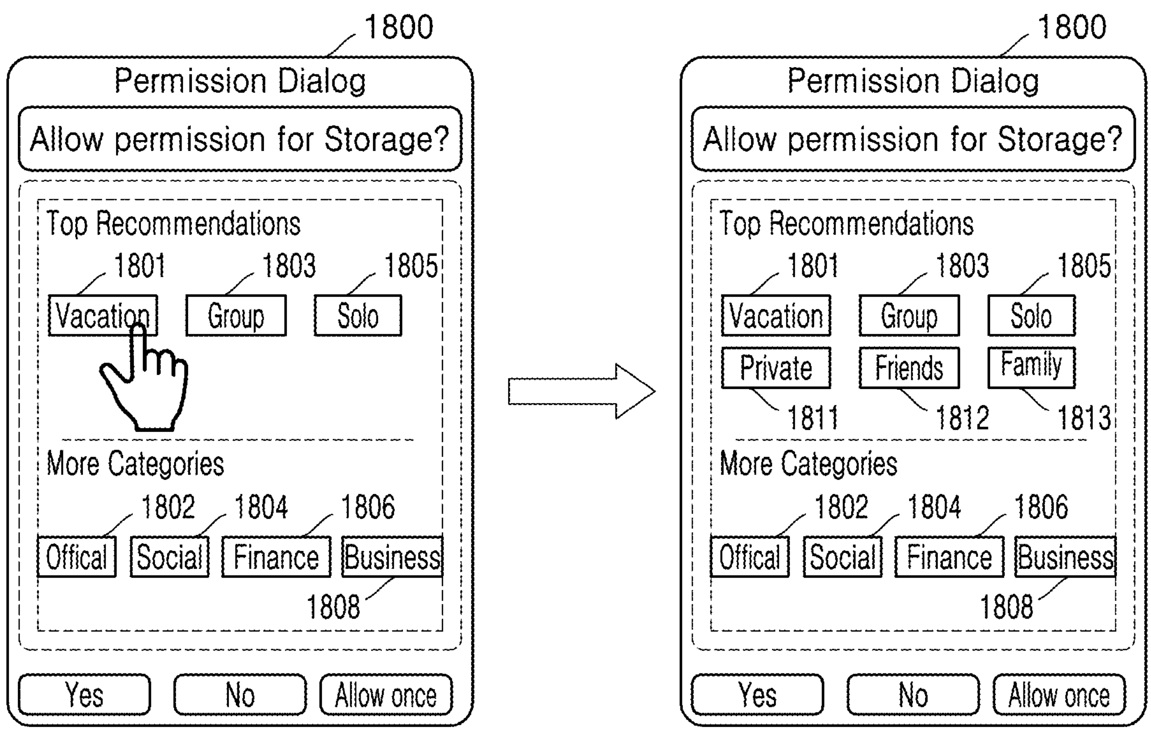

1800
Permission Dialog
Allow permission for Storage?
Top Recommendations
1801
1803
1805
Vacation
Group
Solo
More Categories
1802
1804
1806
Offical
Social
Finance
Business
1808
Yes
No
Allow once
1800
Permission Dialog
Allow permission for Storage?
Top Recommendations
1801
1803
1805
Vacation
Group
Solo
Private
Friends
Family
1811
1812
1813
More Categories
1802
1804
1806
Offical
Social
Finance
Business
1808
Yes
No
Allow once

FIG. 19

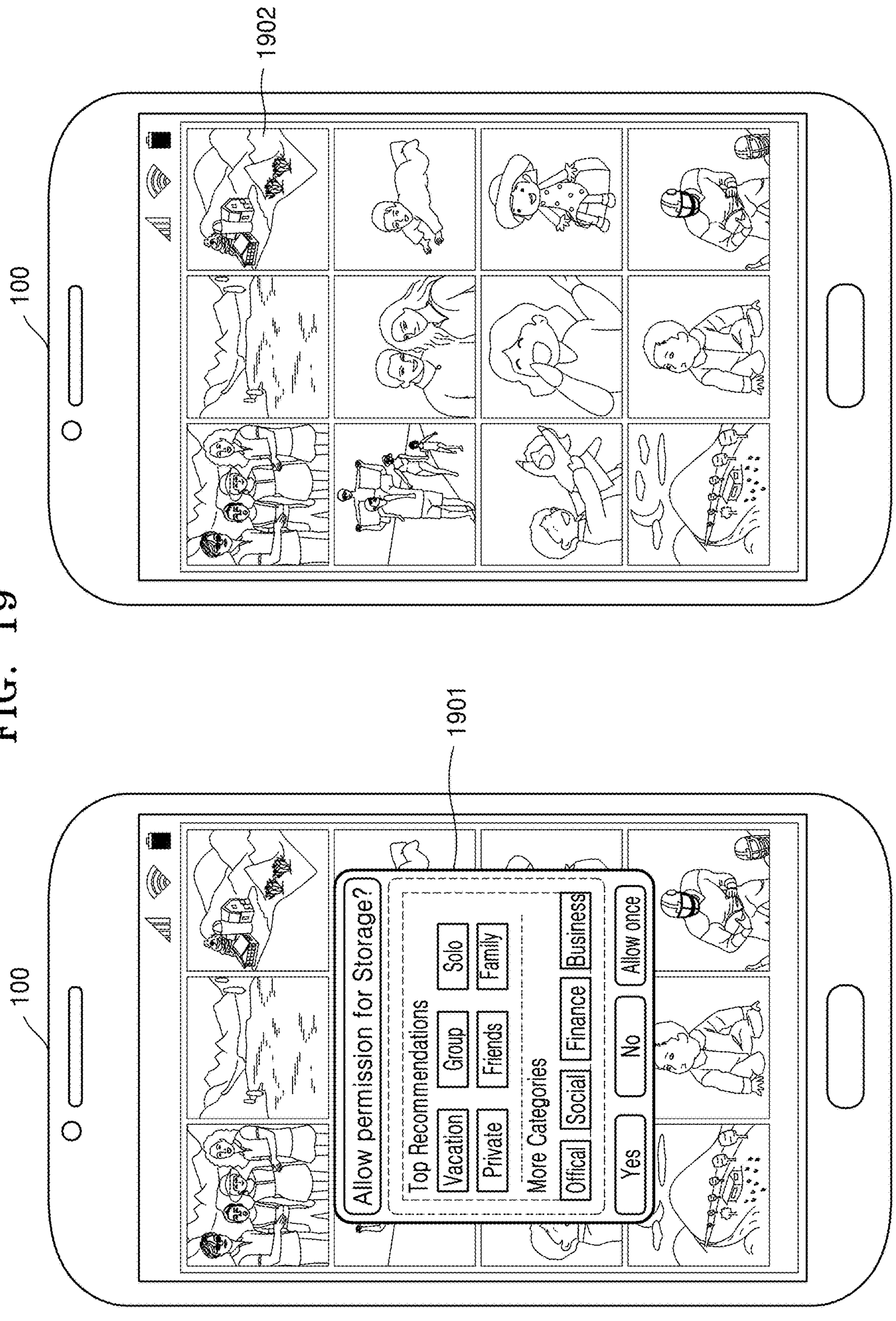

100
1902
100
1901
Allow permission for Storage?
Top Recommendations
Vacation
Group
Solo
Private
Friends
Family
More Categories
Offical
Social
Finance
Business
Yes
No
Allow once

FIG. 20

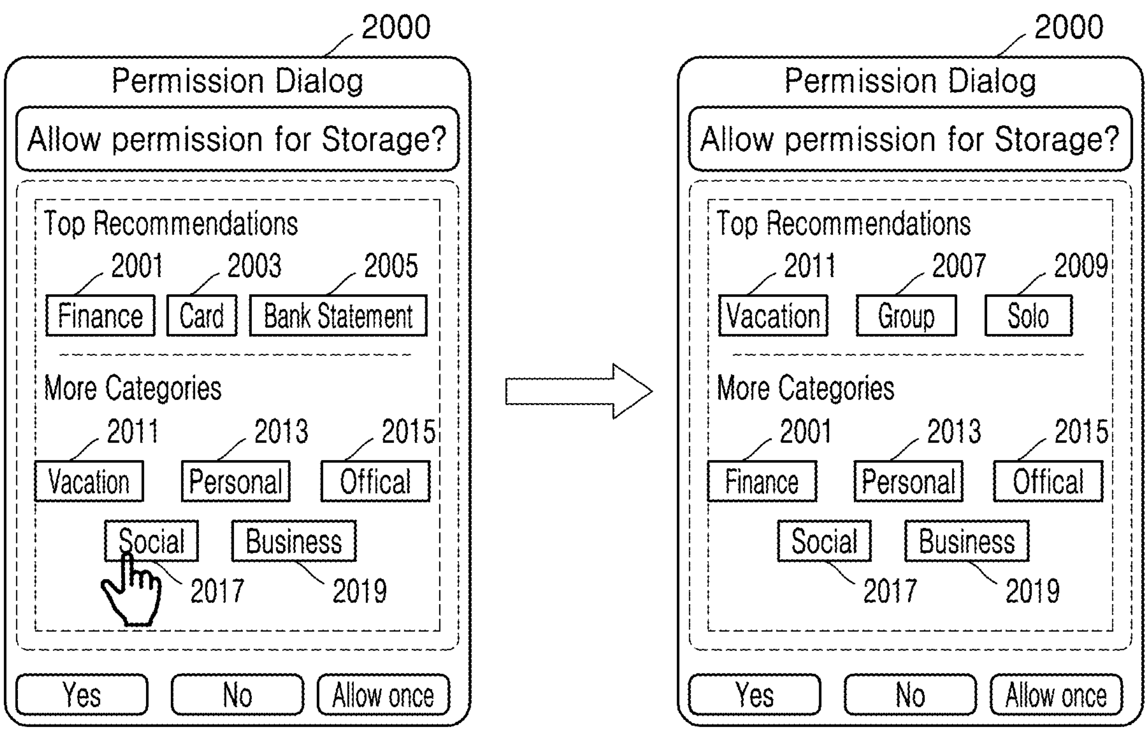
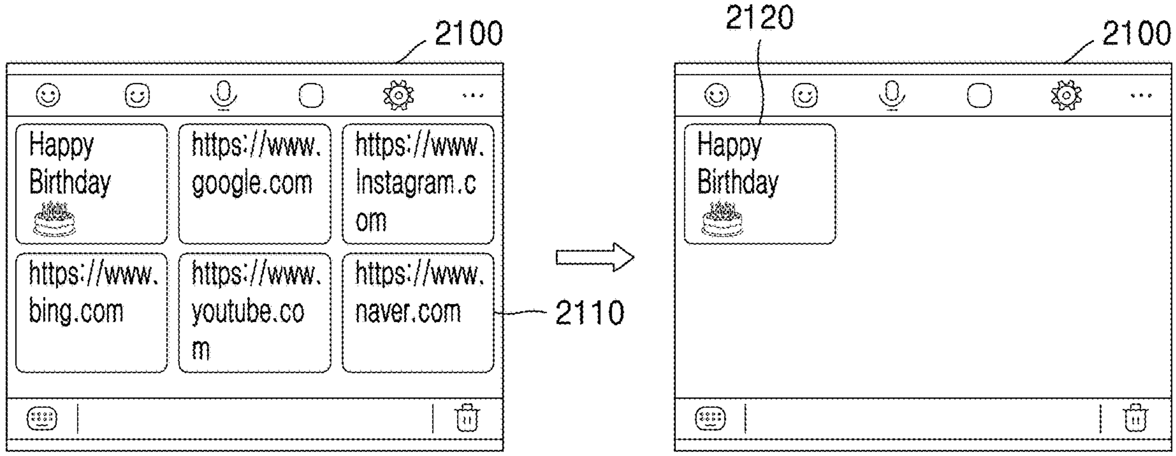

2000
Permission Dialog
Allow permission for Storage?
Top Recommendations
2001
2003
2005
Finance
Card
Bank Statement
More Categories
2011
2013
2015
Vacation
Personal
Offical
Social
Business
2017
2019
Yes
No
Allow once
2000
Permission Dialog
Allow permission for Storage?
Top Recommendations
2011
2007
2009
Vacation
Group
Solo
More Categories
2001
2013
2015
Finance
Personal
Offical
Social
Business
2017
2019
Yes
No
Allow once

FIG. 21


2100
Happy
Birthday
https://www.
google.com
https://www.
Instagram.c
om
https://www.
bing.com
https://www.
youtube.co
m
https://www.
naver.com
2110
2120
2100
Happy
Birthday

ELECTRONIC DEVICE FOR MANAGING ACCESS TO AN APPLICATION AND METHOD THEREOF

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a by-pass continuation application of International Application No. PCT/KR2023/007212, filed on May 25, 2023, which is based on and claims priority to Indian Patent Application number 202241075664, filed on Dec. 26, 2022, in the Indian Patent Office, the disclosures of which are incorporated by reference herein in their entireties.

1. FIELD

[0002] The present disclosure relates to methods and systems for access management in mobile devices and more particularly, to managing access of content and/or sensors for an application in the electronic device.

2. DESCRIPTION OF RELATED ART

[0003] With rapid development in smartphones and associated third-party applications, users are increasingly concerned about privacy and data misuse. Multiple incidents of such blatant misuse of data are reported on a daily basis, which trigger a huge concern regarding privacy among users. Some techniques are implemented to address this problem by introducing more restrictions in allowing data access to installed applications. Operating systems have introduced methods to revoke permissions of applications which are not in use by the user. Such methods have also restricted background access by such applications.

[0004] Systems may not provide permission models for the applications which are more in control of users. Existing techniques may fail to provide protection against application synchronizing private data to a server without user's knowledge. Such a system fails to provide transparency to the user regarding content being accessed by the applications.

[0005] There is therefore a need for a technology capable of facilitating permission handling for applications that may increase privacy protection and may reduce data misuse.

SUMMARY

[0006] According to an aspect of the disclosure, a method for managing access to content for an application of an electronic device includes, receiving, from the application, a content access request for granting access to first data of one or more content types available via the electronic device; displaying one or more labels, from among a plurality of labels, corresponding to the one or more content types; receiving a first user selection of at least one label, from among the one or more labels, corresponding to the one or more content types; and granting, to the application, access to the first data of the one or more content types based on the first user selection.

[0007] The method may further include displaying one or more sub-labels based on the first user selection; receiving a second user selection of at least one sub-label, from among the one or more sub-labels, corresponding to at least one of the one or more content types; and granting, to the application, access to second data of the at least one of the one or more content types based on the second user selection.

[0008] The one or more content types may include at least one of images, videos, text messages, electronic device information, or sensor data.

[0009] The method may further include monitoring a content type of second data accessed by the application; determining whether the content type deviates from the access granted; detecting a spurious access by the application based on the content type deviating from the access granted; and displaying a notification indicating the spurious access.

[0010] The method may further include determining an application type based on metadata associated with the application; and restricting the access to second data from among a set of one or more content types based on the application type.

[0011] The method may further include receiving, from the application, a sensor access request to access one or more sensors of the electronic device; determining second data corresponding to the sensor access request by the application; determining whether the second data complies with a predefined criteria for accessing the one or more sensors; and providing, to the application, access to the one or more sensors based on determining that the second data complies with the predefined criteria.

[0012] The method may further include monitoring the access to the one or more sensors by the application; determining whether the application is in use by a user of the electronic device while the one or more sensors are being accessed by the application; and based on determining the application is not in use, restricting the access to the one or more sensors by the application.

[0013] The method may further include displaying a notification based on the application not being in use by the user while the one or more sensors are being accessed by the application.

[0014] The method may further include generating pseudo content based on the one or more content types; and providing the application access to the pseudo content.

[0015] According to an aspect of the disclosure, an electronic device for managing access to content for an application, the electronic device includes memory storing instructions; and at least one processor; wherein the instructions, when executed by the at least one processor, may cause the electronic device to receive, from the application, a content access request for granting access to first data of one or more content types available via the electronic device; display one or more labels, from among a plurality of labels, corresponding to the one or more content types; receive a first user selection of at least one label, from among the one or more labels, corresponding to the one or more content types; and granting, to the application, access to the first data of the one or more content types based on the first user selection.

[0016] The instructions, when executed by the at least one processor, may cause the electronic device to display one or more sub-labels based on the first user selection; receive a second user selection of at least one sub-label, from among the one or more sub-labels, corresponding to at least one of the one or more content types; and grant, to the application, access to second data of the at least one of the one or more content types based on the second user selection.

[0017] The one or more content types may include at least one of images, videos, text messages, electronic device information, or sensor data.

[0018] The instructions, when executed by the at least one processor, may cause the electronic device to monitor a content type of second data accessed by the application; determine whether the content type deviates from the access granted; detecting a spurious access by the application based on the content type deviating from the access granted; and display a notification indicating the spurious access.

[0019] The instructions, when executed by the at least one processor, may cause the electronic device to determine an application type based on metadata associated with the application; and restrict the access to second data from among a set of one or more content types based on the application type.

[0020] The instructions, when executed by the at least one processor, may cause the electronic device to receive, from the application, a sensor access request to access one or more sensors of the electronic device; determine second data corresponding to the sensor access request by the application; determine whether the second data complies with a predefined criteria for accessing the one or more sensors; and provide, to the application, access to the one or more sensors based on determining that the second data complies with the predefined criteria.

BRIEF DESCRIPTION OF THE DRAWINGS

[0021] The above and other aspects, features, and advantages of certain embodiments of the present disclosure are more apparent from the following description taken in conjunction with the accompanying drawings, in which:

[0022] FIG. **1** illustrates a block diagram of an electronic device implementing access management for an application, according to an embodiment;

[0023] FIG. **2** illustrates a schematic block diagram of the electronic device for managing access to an application, according to an embodiment;

[0024] FIG. **3** illustrates a schematic architecture of the electronic device, including a system for managing access for an application in the electronic device, according to an embodiment;

[0025] FIG. **4** illustrates a block diagram of a content aware privacy module, according to an embodiment;

[0026] FIG. **5** illustrates a flow chart of a method of adding content, according to an embodiment;

[0027] FIG. **6** illustrates a flow chart of a method of querying content by an application, according to an embodiment;

[0028] FIG. **7** illustrates a flow chart of a method of adding content, according to another embodiment;

[0029] FIG. **8** illustrates a flow chart of a method of querying content by an application, according to another embodiment;

[0030] FIG. **9** illustrates a flow chart of a method for permission management of an application, according to an embodiment;

[0031] FIG. **10** illustrates a flow chart of a method of label prediction, according to an embodiment;

[0032] FIG. **11** illustrates a flow chart of a method of label training, according to an embodiment;

[0033] FIG. **12** illustrates a flow chart of a method of anomaly detection, according to an embodiment;

[0034] FIG. **13** illustrates a flow chart of a method for rendering a user interface (UI), according to an embodiment;

[0035] FIG. **14** illustrates a flow chart of a method for cleaning content, according to an embodiment;

[0036] FIG. **15**A illustrates a flow chart of a method for managing access of content to an application in the electronic device, according to an embodiment;

[0037] FIG. **15**B illustrates a flow chart of a method for managing access of content to an application in the electronic device, according to an embodiment;

[0038] FIG. **15**C illustrates a flow chart of a method for managing access of content to an application in the electronic device, according to an embodiment;

[0039] FIG. **16**A illustrates a flow chart of a method for managing access of content for an application in the electronic device, according to another embodiment;

[0040] FIG. **16**B illustrates a flow chart of a method for managing access of content for an application in the electronic device, according to another embodiment;

[0041] FIG. **16**C illustrates a flow chart of a method for managing access of content for an application in the electronic device, according to another embodiment;

[0042] FIG. **16**D illustrates a flow chart of a method for managing access of content for an application in the electronic device, according to another embodiment;

[0043] FIG. **17** illustrates generation of labels for a content access request by a messaging application, according to an embodiment;

[0044] FIG. **18** illustrates recommendation of labels for media access requests by an application, according to an embodiment;

[0045] FIG. **19** illustrates recommendation of labels for media access requests by an application, according to another embodiment;

[0046] FIG. **20** illustrates learning recommendation, according to an embodiment; and

[0047] FIG. **21** illustrates a comparison of clipboard content access according to a technique and an embodiment.

DETAILED DESCRIPTION

[0048] The embodiments described in the disclosure, and the configurations shown in the drawings, are only examples of embodiments, and various modifications may be made without departing from the scope and spirit of the disclosure.

[0049] It will be understood by those skilled in the art that the foregoing description and the following detailed description are explanatory of the disclosure and are not intended to be restrictive thereof.

[0050] The elements shown in the drawings may not have necessarily been drawn to scale. For example, the flow charts illustrate the method in terms of operations involved to help to improve understanding of aspects of the present disclosure. In terms of the construction of the device, one or more components of the device may have been represented in the drawings by symbols, and the drawings may show only those details that are pertinent to understanding the embodiments of the present disclosure so as not to obscure the drawings with details that will be readily apparent to those of ordinary skill in the art having the benefit of the description herein.

[0051] Reference to "an aspect", "another aspect" or similar language means that a feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the present disclosure. Appearances of the phrase "in an embodiment", "in another embodiment" and similar language may, but do not necessarily, all refer to the same embodiment.

[0052] The terms "comprises", "comprising", or any other variations thereof, are intended to cover a non-exclusive inclusion, such that a process or method that comprises a list of operations does not include only those operations but may include other operations not expressly listed or inherent to such process or method. One or more devices or sub-systems or elements or structures or components proceeded by "comprises . . . a" does not, without more constraints, preclude the existence of other devices or other sub-systems or other elements or other structures or other components or additional devices or additional sub-systems or additional elements or additional structures or additional components.

[0053] The expressions "at least one of A, B and C" and "at least one of A, B, or C", both indicate "A", only "B", only "C", both "A and B", both "A and C", both "B and C", and all of "A, B, and C".

[0054] The terms "mobile device", "electronic device" and "smartphone" may be used interchangeably throughout the description.

[0055] To solve the above-mentioned problems, the present disclosure discloses a mechanism to manage access of content and/or sensors by an application installed within a mobile device of the user. The present disclosure enables the user to effectively manage permission of content and sensors for the application.

[0056] According to various embodiments, a method and system for managing access of content for an application in an electronic device is provided. The method includes receiving a request for accessing content by the application, then providing labels associated with the content requested by the user. The method includes receiving user selection of labels and providing access to the application based on the received user selection.

[0057] In some embodiments, the method may also include receiving a sensor access request from the application. In such scenarios, the method includes providing the access to the sensors if the sensor access request complies with a set of predefined rules.

[0058] The technique provided by the present disclosure may facilitate access management for an application to protect user privacy and data.

[0059] FIG. **1** illustrates a block diagram of an electronic device **100** implementing access management for an application **102**, according to an embodiment. Examples of the electronic device **100** may include, but not limited to, smartphones, tablets, laptops, portable computing devices, and so forth. The application **102** may be installed in the electronic device **100**. The application **102** may correspond to a software configured to implement functionality of the electronic device **100**. Examples of the application **102** (may also be referred as "app **102**") may include, but not limited to, a camera application, a messaging application, a web browsing application, an image editing application, a social media application and so forth. In some embodiments, the application **102** may use some data and/or access to one or more sensor(s) of the electronic device **100** to implement the desired functionality. For example, a social media application may request access to image content stored in the electronic device **100** to enable the user to share the stored images over a social media platform. The social media application may request access to the image sensor of the electronic device **100** to enable the user to capture an image and share the captured image over the social media platform.

[0060] To access the content and/or the sensors, the application **102** may generate a request. The application **102** may generate a content access request to access different content types stored in the electronic device **100**. The content types may include, but not limited to, images, videos, text messages, electronic device information, and sensors-related data. The application **102** may generate a sensor access request to access sensor(s) of the electronic device **100**. In an embodiment, each of the content access request and the sensor access request may be sent to a system **104** for approval of such access requests.

[0061] In an embodiment, the system **104** may be installed within the electronic device **100**. In an embodiment, the system **104** may be a part of an operating system of the electronic device **100**. The system **104** may be a standalone entity located remotely to the electronic device **100**. In such condition, the system **104** may be communicably coupled to the electronic device **100** via wireless and/or wired means.

[0062] In an embodiment, the system **104** may be configured to receive the content access request and the sensor access request generated by the application **102**. In an embodiment, the content may be associated with different labels. The system **104** may be configured to store different labels for different content types. In an embodiment, the labels may correspond to names of semantic categories in which the content may be categorized. For example, for images, the labels may include, but not limited to, private and public, or vacation, family, friends and so forth. For messages, the labels may include, but not limited to, financial messages, personal messages, One Time Passwords (OTPs) messages and so forth. In some embodiments, the content may be categorized into semantic categories using on-device Machine Learning (ML) model and/or collaborative learning. In an embodiment, the system **104** may be configured to provide one or more labels from a plurality of labels corresponding to each of content types requested by the application in the content access requests. For example, if the application request access for images. The system **104** may be configured to provide labels corresponding to image media to the user. Such labels may be provided to the user via a user interface (UI) **106** displayed over a display of the electronic device **100**. In an embodiment, the labels corresponding to image media may include, but not limited to, vacation, group, solo, private, friend, family and so forth. If the application **102** requests access to messages, the system **104** may provide labels such as, but not limited to, finance messages, bank accounts, credit cards, and application One Time Passwords (OTP), as shown in FIG. **1**. The system **104** may also provide an option to restrict access to any type of messages or to allow access to all types of messages.

[0063] The system **104** may be configured to receive user selection of label(s) from the UI **106** accessible to the user. For example, for the access content request corresponding to messages, the user may only select labels "credit cards" and "App OTPs", as shown in FIG. **1**.

[0064] The system **104** may be configured to provide content access to the application based on the user selection. The system **104** may restrict access to content for the application **102** which corresponds to labels which are not selected by the user.

[0065] In case of sensor access request, the system **104** may determine whether the received sensor access request complies with a predefined criteria and may only provide access to the sensors upon successfully determining that the sensor access request complies with the predefined criteria. In an embodiment, the predefined criteria may be, but not limited to, a relationship of sensors and different type of application(s), conditions for usage of sensors and so forth. For example, a voice recording application may not be provided access to the camera sensor of the electronic device **100**. In some embodiments, the system **104** may enable a user to select when to allow usage of the sensor by the application **102**. For example, the system **100** may enable the user to allow the application **102** to access the sensors only when the application **102** is in use.

[0066] The electronic device **100** may provide an effective access management technique for the application and protect user privacy and data misuse.

[0067] FIG. **2** illustrates a schematic block diagram of the system **104** for managing access for the application **102** in the electronic device **100**, according to an embodiment. In an embodiment, the system **104** may be included within an electronic device **100** associated with the user. In another embodiment, the system **104** may be configured to operate as a standalone device or a system based in a server/cloud architecture communicably coupled to the electronic device **100**. Examples of the electronic device **100** may include, but not limited to, a mobile phone, a smart watch, a laptop computer, a desktop computer, a Personal Computer (PC), a notebook, a tablet, a mobile phone, and or any other smart device.

[0068] The system **104** may be configured to receive access request(s) from the application **102** and provide access to the application **102** based on user selected criteria and/or predefined criteria. The system **104** may include a processor/controller **202**, an Input/Output (I/O) interface **204**, one or more modules **206**, a transceiver **208**, and a memory **210**.

[0069] In an embodiment, the processor/controller **202** may be operatively coupled to each of the I/O interface **204**, the modules **206**, the transceiver **208** and the memory **210**. In one embodiment, the processor/controller **202** may include at least one data processor for executing processes in Virtual Storage Area Network. The processor/controller **202** may include processing units, such as integrated system (bus) controllers, memory management control units, floating point units, graphics processing units, digital signal processing units, for example. In one embodiment, the processor/controller **202** may include a central processing unit (CPU), a graphics processing unit (GPU), or both. The processor/controller **202** may be one or more general processors, digital signal processors, application-specific integrated circuits, field-programmable gate arrays, servers, networks, digital circuits, analog circuits, combinations thereof, or other now known or later developed devices for analyzing and processing data. The processor/controller **202** may execute a software program, such as code generated manually (for example, programmed) to perform the desired operation.

[0070] The processor/controller **202** may be disposed in communication with one or more input/output (I/O) devices via the I/O interface **204**. The I/O interface **204** may employ communication code-division multiple access (CDMA), high-speed packet access (HSPA+), global system for mobile communications (GSM), long-term evolution (LTE), WiMax, or the like.

[0071] Using the I/O interface **204**, the system **104** may communicate with one or more I/O devices, the electronic devices associated with the human-to-human conversation. For example, the input device may be an antenna, microphone, touch screen, touchpad, storage device, transceiver, or video device/source, for example. The output devices may be a printer, fax machine, video display (e.g., cathode ray tube (CRT), liquid crystal display (LCD), light-emitting diode (LED), plasma, Plasma Display Panel (PDP), Organic light-emitting diode display (OLED) or the like), or audio speaker, for example. In an embodiment, the system **104** may communicate with the electronic device associated with the user using the I/O interface **204**.

[0072] The processor/controller **202** may be disposed in communication with a communication network via a network interface. In an embodiment, the network interface may be the I/O interface **204**. The network interface may connect to the communication network to enable connection of the system **104** with the outside environment and/or device/system. The network interface may employ connection protocols including, without limitation, direct connect, Ethernet (e.g., twisted pair 10/100/1000 Base T), transmission control protocol/internet protocol (TCP/IP), token ring, or IEEE 802.11a/b/g/n/x, for example. The communication network may include, without limitation, a direct interconnection, local area network (LAN), wide area network (WAN), wireless network (e.g., using Wireless Application Protocol), or the Internet, for example. Using the network interface and the communication network, the voice assistant device **201** may communicate with other devices. The network interface may employ connection protocols including, but not limited to, direct connect, Ethernet (e.g., twisted pair 10/100/1000 Base T), transmission control protocol/internet protocol (TCP/IP), token ring, or IEEE 802.11a/b/g/n/x, for example.

[0073] In an embodiment, the processor/controller **202** may receive the content access request from the application **102** to access one or more content types available at the electronic device **100**. The processor/controller **202** may be configured to provide one or more labels corresponding to each one of the one or more content types requested by the application to a user of the electronic device **100**. In an embodiment, the labels may correspond to semantic categories associated with content types. The semantic categories may be based on demographic profile of the user. The demographic profile of the user may be based on information such as, but not limited to, age, sex, race and user-interest. In some embodiments, a privacy profile corresponding to the user may be generated based on the demographic profile information of the user. The privacy profile may indicate user preference with respect to sharing of data and/or sensor access with different type of applications. In some embodiments, the content may be categorized into the semantic categories based on the user's privacy profile. In an embodiment, the processor/controller **202** may be configured to determine request related data corresponding to the content access request. The request related data may include information such as, but not limited to, a type of the application, a use of requested content by the application, an information associated with access provided to the application based on one or more previous requests, or an information associated with access provided to one or more applications similar to the application. The processor/controller **202** may provide the labels based on the determined request related data.

[0074] The processor/controller **202** may also be configured to receive a user selection of at least one label from the one or more labels corresponding to the one or more content types via the I/O interface **204**. The processor/controller **202** may be configured to provide an access to the application **102** to the one or more content types requested by the application based on the user selection of label(s).

[0075] In some embodiments, the processor/controller **202** may be configured to provide one or more sub-labels based on the user selection of at least one label. For example, in case the application **102** requested access to images. Initially, the processor/controller **202** may provide labels such as, vacation, public, private or family and once the user selects a label, the processor/controller **202** may provide a set of sub-labels corresponding to the user selected label. For example, if a user selects vacation, the processor/controller **202** may provide sub-labels such as, official, social, solo, family, friends outing, international, and so forth. The processor/controller **202** may also be configured to receive a user selection of at least one sub-label from the presented sub-labels. The processor/controller may be configured to provide access to the application **102** to the one or more content types requested by the application **102** based on the user selection of the sub-label(s).

[0076] In some embodiments, the processor/controller **202** may be configured to monitor an actual content type access by the application **102**. The processor/controller **202** may be configured to determine whether the actual content type is different from the provided access to the content type. In an example scenario, the application **102** may be provided with access to images categorized under the label "vacation". Now, if the application **102** tries to access images categorized under labels other than "vacation", such as, private or family, the processor/controller **202** may consider such access by the application **102** as a spurious access. The processor/controller **202** may generate an alert for the user and display the determined spurious access by the application **102**. The processor/controller **202** may enable the user to take appropriate measures to stop spurious access by the application **102**. For example, the user may uninstall the application **102** in case of determination of the spurious access by the application **102**.

[0077] In some embodiments, the processor/controller **202** may be configured to determine a type of the application based on metadata associated with the application **102**. Examples of different types of application may include, but not limited to, a gaming application, a social media application, a financial application, and so forth. The processor/controller **202** may be configured to restrict access to a set of content type for the application **102** based on the type of the application. For example, if the type of application is the social media application, the processor/controller **202** may restrict access to financial messages. The processor/controller **202** may generate a suggestion for the user to restrict access to content type based on the determined type of application.

[0078] In some other embodiments, the processor/controller **202** may be configured to receive a sensor access request from the application **102**. The sensor access request may correspond to a request to access sensor(s) of the electronic device **100** by the application **102**. The processor/controller **202** may be configured to determine sensor related data corresponding to the sensor access request by the application **102**. The sensor related data may include information, such as, but not limited to, a type of sensor, an application of the sensor, and usage timing of the sensor and so forth. For example, the sensor related data may indicate that the application **102** intends to access a camera sensor (for example, the type of sensor) to capture an image (for example, the application of the sensor) while the user is accessing the application **102** (for example, usage timing of the sensor). The processor/controller **202** may be configured to determine whether the sensor related data complies with a predefined criteria to access the sensors or not. The predefined criteria may define rules relating to usage timing of sensors, access to sensors based on a type of application and associated application, and so forth. The predefined criteria may be defined by the system **104** or the user based on the suggestion from the system **104**. Upon determining that the sensor request related data complies with the predefined criteria the processor/controller **202** may provide access to the requested sensors.

[0079] In an embodiment, the processor/controller **202** may be configured to monitor the access of the sensors by the application **102**. The processor/controller **202** may be configured to determine whether the application **102** is in use by the user while accessing the one or more sensors and may restrict the access to the sensors by the application **102** upon determining that the application **102** is not in use by the user. In some embodiments, the processor/controller **202** may be configured to generate an alert for the user upon determining that the application is not in use while accessing the sensors. The alert may be provided to the user via the I/O interface **204** to enable the user to take appropriate measures to prevent such restricted access. The processor/controller **202** may protect user privacy and enhance protection against data misuse.

[0080] In an embodiment, the processor/controller **202** may also be configured to generate pseudo content based on the one or more content types requested by the application **102**. The processor/controller **202** may provide access to such pseudo content when the application **102** requests for content. The processor/controller **202** enables a user to effectively test functionality of the application **102**. By providing access to the pseudo content, the processor/controller **202** may enable the application **102** to operate and thereby the processor/controller **202** may implement effective monitoring of spurious access by the application **102**.

[0081] In some embodiments, the memory **210** may be communicatively coupled to the at least one processor/controller **202**. The memory **210** may be configured to store data, instructions executable by the at least one processor/controller **202**. In one embodiment, the memory **210** may communicate via a bus within the system **104**. The memory **210** may include, but not limited to, a non-transitory computer-readable storage media, such as various types of volatile and non-volatile storage media including, but not limited to, random access memory, read-only memory, programmable read-only memory, electrically programmable read-only memory, electrically erasable read-only memory, flash memory, magnetic tape or disk, optical media and the like. In one example, the memory **210** may include a cache or random-access memory for the processor/controller **202**. The memory **210** may be separate from the processor/controller **202**, such as a cache memory of a processor, the system memory, or other memory. The memory **210** may be an external storage device or database for storing data. The memory **210** may be operable to store instructions executable by the processor/controller **202**. The functions, acts or tasks illustrated in the drawings or described may be performed by the programmed processor/controller **202** for executing the instructions stored in the memory **210**. The functions, acts or tasks are independent of the particular type of instructions set, storage media, processor or processing strategy and may be performed by software, hardware, integrated circuits, firmware, micro-code and the like, operating alone or in combination. Likewise, processing strategies may include multiprocessing, multitasking, parallel processing, and the like.

[0082] In some embodiments, the modules **206** may be included within the memory **210**. The memory **210** may further include a database **212** to store data. The one or more modules **206** may include a set of instructions that may be executed to cause the system **104** to perform any one or more of the methods/processes disclosed herein. The one or more modules **206** may be configured to perform the operations using the data stored in the database **212**, to determine semantic points in a human-to-human conversation as discussed herein. In an embodiment, each of the one or more modules **206** may be a hardware unit which may be outside the memory **210**. The memory **210** may include an operating system **214** for performing one or more tasks of the system **104**, as performed by an operating system in the communications domain. The transceiver **208** may be configured to receive and/or transmit signals to and from the electronic device associated with the user. In one embodiment, the database **212** may be configured to store the information by the one or more modules **206** and the processor/controller **202** to perform one or more functions for determining semantic points in a human-to-human conversation.

[0083] In an embodiment, the I/O interface **204** may enable input and output to and from the system **104** using devices such as, but not limited to, display, keyboard, mouse, touch screen, microphone, speaker and so forth.

[0084] The disclosure includes a non-transitory computer-readable medium that includes instructions or receives and executes instructions responsive to a propagated signal. The instructions may be transmitted or received over the network via a communication port or interface or using a bus. The communication port or interface may be a part of the processor/controller **202** or may be a separate component. The communication port may be created in software or may be a physical connection in hardware. The communication port may be configured to connect with a network, external media, the display, or any other components in the system, or combinations thereof. The connection with the network may be a physical connection, such as a wired Ethernet connection or may be established wirelessly. Likewise, the additional connections with other components of the system **104** may be physical or may be established wirelessly. The network may be directly connected to the bus.

[0085] FIG. **3** illustrates a schematic architecture of the electronic device **100** including a system **104** for managing access for an application in the electronic device, according to an embodiment. FIG. **4** illustrates a block diagram of a content aware privacy module, according to an embodiment. FIG. **3** may be explained in conjunction with FIG. **4**.

[0086] In one embodiment, the system **104** may receive an access request from the application **102**. The application **102** may include applications **302** and **304**. The applications **302** may include system applications and user-installed applications. For example, the applications **302** may include a camera application, a message application, a Social Network Service (SNS) application, and so forth.

[0087] In an embodiment, the applications **302** and **304** may be communicably coupled with the modules **206** via an interface which may be configured to receive and process the access request from the applications and share with a content aware privacy module **305**.

[0088] In an embodiment, the system **104** may include the content aware privacy module **305** (interchangeably referred as "the module **305**"). The module **305** may be a part of the modules **206**. The module **305** may be configured to receive access request(s) from the application **102**. The module **305** may further include a content aware provider module **306**, a label recommender module **308**, a user dashboard module **310**, an anomaly detector module **312**, a content trash policy module **314**, a permission manager module **316**, a collaborative clustering module **318**, a cluster manager module **320** and an access logger module **322**. Each of modules **306-322** may be communicably coupled with each other.

[0089] The content aware provider module **306** may be configured to receive the access request(s) and determine request-related data associated with the access request. In an embodiment, the request-related data may include a type of the application, a use of requested content by the application, an information associated with access provided to the application based on one or more previous requests, or an information associated with access provided to one or more applications similar to the application. The content aware provider module **306** may also be configured to generate demographic profile of the user. The content aware provider module **306** may be configured to provide label(s) corresponding to content types requested by the application via the access request based on the request-related data and demographic profile of the user. In some embodiments, the labels may correspond to semantic classes corresponding to the content. In some embodiments, the content aware provider module **306** may include different Application Programming Interfaces (APIs) to implement various functionality of the content aware provider module **306**. The different APIs may establish communication interface between the application **102** and the content aware provider module **306**. In some embodiments, the APIs may define rules to enable communication between the application **102** and the content aware provider module **306** using requests and responses.

[0090] In an embodiment, the label recommender module **308** may be configured to recommend semantic classes based on request related data and the demographic profile of the user. In some embodiments, the label recommender module **308** may also be configured to recommend sub-labels (for example, sub-categories and/or related classes) based on the recommended semantic classes and corresponding user selection. The label recommender module **308** may include an application category identifies configured to identify a type of the application **102**. The label recommender module may store the user demographic profiles. In some embodiments, the label recommender module **308** may be configured to recommend labels corresponding to the semantic classes of the content based on the type of the application **102** and the demographic profile of the user.

[0091] The user dashboard module **310** may be configured to generate user interfaces (UIs) for the user to display labels and/or sub-labels and enable the user to view or modify user selection of the labels and sub-labels. The user dashboard module **310** may be configured to enable the user to view and manage access permission granted to the application **102**. In some embodiments, the user dashboard module **310** may be configured to display and manage data access logs corresponding to the application **102**, anomalies in access by the application **102** and trashing of the data. The user dashboard module **310** may be configured to generate a dashboard UI to enable the user to view and manage permission corresponding to different application installed within the electronic device **100**. The user dashboard module **310** may be configured to generate permission UI enabling the user to grant/reject permission requests from the application **102**.

[0092] The anomaly detector module **312** may be configured to detect anomalies in content access by the application **102**. The anomaly detector module **312** may be configured to store the content access permission granted to the application **102**. The anomaly detector module **312** may be configured to monitor an actual content type accessed by the application. The anomaly detector module **312** may be configured to monitor unauthorized attempt to access content. The anomaly detector module **312** may be configured to determine whether the actual content type is different than the access provided to the application **102** by the user. Upon determining that the content type, the application **102** has tried access for or has gained access for, is not authorized by the user, the anomaly detector module **312** may consider such access as anomaly or a spurious access. In some embodiments, the anomaly detector module **312** may be configured to generate an alert upon determining such anomaly or spurious access by the application **102**. In an embodiment, the anomaly detector module **312** may include a detector configured to detect anomalies in content access by the application **102**, an anomaly detection receiver configured to receive the detected anomalies and a notifier configured to generate an alert for the user based on the received anomalies.

[0093] The content trash policy module **314** may be configured to determine and/or store polices to delete content from the electronic device **100**. The policies may define rules for deletion of content based on labels, time period, usage and so forth. In an embodiment, the content trash policy module **314** may include a content trash module configured to delete the content based on the stored policies. In some other embodiments, the content trash module may delete the content based on inputs received from the user. The content trash policy module **314** may include trash policy determiner configured to generate and update policies to effectively delete the content from the electronic device **100**.

[0094] The permission manager module **316** may be configured to configured to manage permission granted to the application **102** along with user preferences. In some embodiments, the permission manager module **316** may be configured to provide backend support for the user dashboard module **310**.

[0095] The collaborative clustering module **318** may be configured to train and update the label recommender module **308** based on user-related data corresponding to different users. In an embodiment, the different users may share a similar demographic profile. The collaborative clustering module **318** may be configured to implement distributed learning technique to enable the label recommender module **308** to learn and update labels corresponding to different content types. In an embodiment, the collaborative clustering module **318** may include clustering model configured to generate cluster of content based on corresponding labels. The collaborative clustering module **318** may include label recommender model configured to train and update the label recommender module **308** based on user-related data corresponding to different users.

[0096] In an embodiment, the collaborative clustering module **318** may also include a Machine Learning (ML) framework includes Federated Learning (FL) APIs and learning APIS. The collaborative clustering module **318** may be configured train and update the label recommender module **308** using the ML framework. In some embodiment, the collaborative clustering module **318** may be coupled with a server (now shown) to implement the training process. The server may be located remotely to the electronic device. The server may include a round management unit configured to execute training rounds for the collaborative clustering module **318**. The server may include model store configured to store various model used for training the collaborative clustering module **318**. The server may include client manger configured to manage various electronic devices connected to the server. The server may also include an aggregator configured to aggregates model weights received from various electronic devices. The server may further include a policy manager configured to manage policies for training and aggregation. The policies may include information such as, but not limited to, selection criteria of model, selection timeout, aggregation strategy and so forth. The server may include training plan store configured to store training paths with corresponding policy information. In an embodiment, the server may receive various training models from the different electronic devices connected with the server. The server may aggregates the models and shared an updated model with the connected electronic devices.

[0097] The cluster manager module **320** may be configured to manager cluster of different content types and corresponding labels. The cluster manager module **320** may be configured to maintain a database for content and label mapping. The cluster manager module **320** may create a relationship between content and corresponding label by generating and mapping content IDs with corresponding content labels. In some embodiments, the cluster manager module **320** may be configured to perform query expansion to determine various parameters in access request received from the application **102**. The cluster manager module **320** may include a content mapper configured to map content with corresponding labels.

[0098] The access logger module **322** may be configured to log all the access requests or accesses performed by the application **102**. The access logger module **322** may be configured to provide data to the anomaly detector module **312** to enable the module **312** detect anomalies. In an embodiment, the access logger module **322** may include action monitor configured to monitor accesses by the application **102**. The access logger module **322** may include an anomaly detection listener configured to receive anomalies detected by the anomaly detector module **312**.

[0099] The system **104** may also include a device content database **324**. The device content database **324** may be stored with the memory **210**. The device content database **324** may be configured to store media content corresponding to various content types in the electronic device **100**. The content types may include, but not limited to, media content, messages, clipboard and so forth.

[0100] The system **104** may also include a content aware privacy engine (CAPE) database **326** (interchangeably referred as "the database **326**"). The database **326** may be configured to store content/information such as, but not limited to, pseudo content, application permission, cluster labels, content access logs and content mapping. The system **104** may include an encrypted storage **328** configured to store files corresponding to different content types.

[0101] FIG. **5** illustrates a flow chart of a method **500** of adding content, according to an embodiment. The method **500** corresponds to adding new content to a database of the electronic device **100**. In some embodiments, the new content/data may be received from the application **102**.

[0102] At operation **502**, the method **500** includes classifying new data into higher level labels. In an embodiment, the system **104** may be configured to implement machine learning technique to classify the data into higher level labels. At operation **504**, the method **500** includes updating the cluster information to include the added data. Next at operation **506**, the method **500** includes generating secondary labels for the newly added data. The method **500** includes utilizing clustering machine learning models to generate secondary level of semantic categories/labels for the newly added data. For example, if the added data corresponds to financial data, a higher-level label may be "banking" and the secondary level of semantic categories may include "accounts", "credits", and "insurance". At operation **508**, the method **500** includes determining whether the generated secondary label exists or not. Upon determining that the generated secondary label does not exist, the method **500** may move to operation **510**. At operation **510**, the method **500** includes storing the label information corresponding to the secondary label and update the cluster database. At operation **512**, the method **500** includes updating the server to implement collaborative learning. In case a new label or sub-label is determined, the method include 500 updating the server with the newly determined labels or sub-labels.

[0103] FIG. **6** illustrates a flow chart of a method **600** of querying content by an application, according to an embodiment. The method **600** may initialize when the application request for content corresponding a content label.

[0104] At operation **602**, the method **600** includes check for permission corresponding the requested content label. At operation **604**, the method **600** includes determining whether the application **102** is granted with permission to access the content with the requested label or not. Upon determining that the permission corresponding to the requested label has not been granted to the application **102**, the method **600** moves to operation **606** and displays a user interface (UI) to the user indicating that the application **102** has requested for permission to access data corresponding to the requested content label. The UI may also enable the user select whether the user wishes to allow the application **102** access to the requested content label or not. Upon determining that the permission corresponding to the requested has already been granted to the application **102**, the method **600** moves to operation **608** and determines whether the granted permission corresponds to pseudo content or actual content. Upon determining that the granted permission corresponds to pseudo content, the method **600** moves to operation **616**. At operation **616**, the method **600** includes querying the cluster manager for pseudo content for the request content label. At operation **618**, the method **600** includes returning the data to the application.

[0105] Further at operation **608**, upon determining the granted permission is not for pseudo content but for actual content, the method **600** moves to operation **610**. At operation **610**, the method **600** includes querying the cluster manager for data corresponding to the requested content label. At operation **612**, the cluster manager module yield data matching with the requested content label. At operation **614**, the method **600** includes returning the data to the application **102**.

[0106] FIG. **7** illustrates a flow chart of a method **700** of adding content, according to another embodiment.

[0107] At operation **702**, the method **700** includes receiving label and content. In an embodiment, the content may be received from the application **102** and the corresponding label may be received from the content aware provider module **306**. At operation **704**, the method **700** includes adding the received content to the device content database **324** based on the received label. At operation **706**, the method **700** includes storing the label and content mapping IDs into the cluster database.

[0108] FIG. **8** illustrates a flow chart of a method **800** of querying content by an application, according to another embodiment.

[0109] At operation **802**, the method **800** includes receiving label and query parameters. In an embodiment, the query parameters may correspond to parameters associated with a query for content received from the application **102**. The cluster manager module **320** may add the label to the query based on the query parameters. At operation **804**, the method **800** includes querying the database for content IDs corresponding the requested label. At operation **806**, the method **800** includes fetching data from the device content database **324** with the fetched content IDs. At operation **808**, the method **800** includes storing access log into logger database. At operation **810**, the method **800** includes returning the data to the application **102**.

[0110] FIG. **9** illustrates a flow chart of a method **900** for permission management of an application, according to an embodiment.

[0111] At operation **902**, the application **102** requests labels corresponding to content from the system **104**. At operation **904**, the cluster manager module **320** of the system **104** return higher level labels corresponding to the content. At operation **906**, the application **102** requests the permission manager module **316** to display UI for the labels provided by the cluster manager module **320**. At operation **908**, the label recommender module **308** predicts labels to be selected. In an embodiment, the label recommender module **308** may predicts the labels to be selected by the user based on a type of application **102** and user's demography profile. At operation **910**, the permission manager module **316** displays the user with the labels selected by the label recommender module **308**. At operation **912**, the user selects labels to modify the label selection. In an embodiment, the user may continue with labels selected by the label recommender module **308**. In another embodiment, the user may modify the label selection. At operation **914**, the system **104** saves cluster access information for the application **102** into the application permission database based on the user selection for further access requests. In some alternative embodiments, the application **102** may directly requests the permission manager module **316** to display user with labels to get access to request content.

[0112] FIG. **10** illustrates a flow chart of a method **1000** of label prediction, according to an embodiment. At operation **1002**, the permission manager module **316** requests the label recommender module **308** for prediction of labels corresponding to content access request received from the application **102**. At operation **1004**, the label recommender module **308** uses pre-trained models to predict labels corresponding to the requested content. At operation **1006**, the label recommender module **308** returns the predicted labels.

[0113] FIG. **11** illustrates a flow chart of a method **1100** of label training, according to an embodiment.

[0114] At operation **1102**, the user modifies label selection suggested by the label recommender module **308**. At operation **1104**, the label recommender module **308** learns the label selection by the user and updates a distributed model used to predict labels. At operation **1106**, FL training round sends model updates to the server for aggregation. Model updates with corresponding model weights may be transmitted to the server for aggregation.

[0115] FIG. **12** illustrates a flow chart of a method **1200** of anomaly detection, according to an embodiment.

[0116] In an embodiment, the method **1200** initiate with a content access request received from the application **102**. At operation **1202**, the method **1200** includes logging content access request(s) received from the application **102**. At operation **1204**, the method **1200** includes determining whether the user has granted permission to the application **102** or not. Upon determining that the application **102** has not been granted permission to access the content, the method **1200** moves to operation **1206**. At operation **1206**, the method **1200** includes display permission request UI to the user. Upon determining that the application **102** has been granted permission to access the content, the method **1200** moves to operation **1208**. At operation **1208**, the method **1200** includes monitoring content access by the application **102**. Based on the content accessed by the application, the method **1200** includes determining anomalies in content access, as shown by the operation **1210**. The different kind of anomalies which may be detected by the system **104** may include, but not limited to, content category request anomaly, background access anomaly, frequent access anomaly, data upload anomaly and so forth. The content category request anomaly may be detected when the application **102** fetches and stores data from non-permitted label/class. The background access anomaly may be detected when the application **102** fetches and stored data during the inactive state or when the application is in background. The frequent access anomaly may be detected when the application **102** makes abnormal number of requests to access the data. The data upload anomaly may be detected when the application **102** tries to upload data without user's consent and/or knowledge. At operation **1212**, the method **1200** includes determining whether the anomaly has been reported or not. In case, no anomaly has been detected, the method **1200** moves back to operation **1208**. In case an anomaly is detected, the method **1200** moves to operation **1214**. At operation **1214**, the method **1200** includes alerting the user based on reported anomaly. In some embodiments, the anomaly situations may be flagged off using FL model if another user with similar demography profile ratifies the situation.

[0117] FIG. **13** illustrates a flow chart of a method **1300** for rendering a user interface (UI), according to an embodiment. At operation **1302**, the method **1300** includes fetching all the labels from the cluster manager module. At operation **1304**, the method **1300** includes fetching access requests by the application **102** for each label. At operation **3106**, the method **1300** includes fetching label wise permission ordered by the application **102**. At operation **1308**, the method **1300** includes fetching all anomalies detected by the system **104**. At operation **1310**, the method **1300** includes rendering information in UI.

[0118] FIG. **14** illustrates a flow chart of a method **1400** for cleaning content, according to an embodiment.

[0119] At operation **1402**, the method **1400** includes fetching labels from the cluster manager module. At operation **1404**, the method **1400** includes rendering labels for user selection. At operation **1406**, the method **1400** includes fetching user selection. At operation **1408**, the method includes requesting selected labels content deletion. At operation **1410**, the cluster manager deletes the content for selected labels.

[0120] FIGS. **15**A-**15**C illustrate a flow chart of a method **1500** for managing access of content for an application in an electronic device, according to an embodiment.

[0121] At operation **1502**, the method **1500** includes receiving a content access request from the application **102**. The content access request may correspond to a request to access one or more content types available at the electronic device **100**. At operation **1504**, the method **1500** includes providing one or more labels from a plurality of labels corresponding to each one of the one or more content types requested by the application to a user of the electronic device **100**. At operation **1506**, the method **1500** includes receiving a user selection of at least one label from the one or more labels corresponding to the one or more content types. Next at operation **1508**, the method **1500** includes providing, one or more sub-labels based on the user selection of the at least one label. At operation **1510**, the method **1500** includes receiving a user selection of at least one sub-label from the one or more sub-labels corresponding to the one or more content types. At operation **1512**, the method **1500** includes providing, to the application, an access to the one or more content types requested by the application based on the user selection.

[0122] Further at operation **1514**, the method **1500** includes monitoring an actual content type accessed by the application. At operation **1516**, the method **1500** includes determining whether the actual content type is different from the provided access to the one or more content types. Next at operation **1518**, the method **1500** includes determining a spurious access by the application to alert the user in response to determining that the actual content type accessed by the application is different than the provided access to the one or more content types. At operation **1520**, the method **1500** includes displaying the determined spurious access by the application **102**.

[0123] At operation **1522**, the method **1500** includes determining a type of the application based on one or more metadata associated with the application. Next at operation **1524**, the method **1500** includes restricting access to a set of content type for the application based on the type of application.

[0124] At operation **1526**, the method **1500** includes receiving, from the application, a sensor access request to access one or more sensors of the electronic device. At operation **1528**, the method **1500** includes determining sensor request related data corresponding to the sensor access request by the application. At operation **1530**, the method **1500** includes determining whether the sensor request related data complies with a predefined criteria to access the one or more sensors. At operation **1532**, the method **1500** includes providing, to the application, an access to the one or more sensors requested by the application upon determining that the sensor request related data complies with the predefined criteria to access the one or more sensors.

[0125] At operation **1534**, the method **1500** includes monitoring the access of the one or more sensors by the application. At operation **1536**, the method **1500** includes determining whether the application is in use by the user while accessing the one or more sensors. At operation **1538**, the method **1500** includes restricting the access to the one or more sensors by the application in response to determining that the application is not in use by the user while accessing the one or more sensors.

[0126] At operation **1540**, the method **1500** includes generating an alert for the user upon determining the application is not in use by the user while accessing the one or more sensors. At operation **1542**, the method **1500** includes generating pseudo content based on the one or more content type requested by the application. At operation **1544**, the method **1500** includes providing access to the pseudo content to the application.

[0127] FIGS. **16**A-**16**D illustrate a flow chart of a method **1600** for managing access of content for an application in an electronic device, according to another embodiment.

[0128] At operation **1602**, the method **1600** includes receiving, from the application, a content access request to access one or more content types available at the electronic device. At operation **1604**, the method **1600** includes determining request-related data corresponding to the content access request to access by the application. At operation **1606**, the method **1600** includes determining a plurality of labels associated with each of the one or more content types requested by the application. At operation **1608**, the method **1600** includes providing one or more labels from the plurality of labels corresponding to each one of the one or more content types requested by the application to a user of the electronic device, based on the determined request-related data. At operation **1610**, the method **1600** includes receiving a user selection of at least one label from the one or more labels corresponding to the one or more content types. At operation **1611**, the method **1600** includes providing, one or more sub-labels based on the user selection of the at least one label. At operation **1612**, the method **1600** includes receiving a user selection of at least one sub-label from the one or more sub-labels corresponding to the one or more content types. At operation **1614**, the method **1600** includes providing, to the application, an access to the one or more content types requested by the application based on the user selection.

[0129] At operation **1616**, the method **1600** includes learning user preferences based on user selections of the at least one label for one or more access requests for one or more content types by the application. At operation **1618**, the method **1600** includes updating the one or more semantic categories for the one or more content types based on the user preferences.

[0130] At operation **1620**, the method **1600** includes determining a rank associated with each of the plurality of labels associated with each of the one or more content types requested by the application based on the learned user preferences. At operation **1622**, the method **1600** includes providing the one or more labels associated corresponding to each one of the one or more content types based on the determined rank corresponding to each label for further recommendations.

[0131] For additional implementation details of operations **1624-1654**, reference may be made to the descriptions of operations **1514-1544**.

[0132] FIG. **17** illustrates generation of labels for a content access request by a messaging application, according to an embodiment.

[0133] In the illustrated embodiment, the messaging application **1700** requests access to messages stored in the electronic device **100**. The messaging application **1700** requests to select a grant permission button **1701**. The system **104** may generate labels corresponding to the grant permission button **1701** and display to the user via a user interface **1702**. In an embodiment, the labels include finance, OTP, credit and debit. The system **104** may also generate some sub-labels such as, offers, personal, official, spam and business. The UI **1702** may be configured to receive user selection of one or more labels and/or sub-label to allow the messaging application to access the user selected messages. The UI **1702** may also enable the user to reject the request or allow the request for one time.

[0134] FIG. **18** illustrates recommendation of labels for media access request by an application, according to an embodiment.

[0135] In the illustrated embodiment, the application requests access for storage for access image file. The system **104** may display labels **1800** based on received request. For example, the system **104** may display recommendations such as "vacation" **1801**, "group" **1803**, and "solo" **1805**. The system **104** may display additional recommendations such as "official" **1802**, "social" **1804**, "finance" **1806** and "business" **1808**. Once a user selects a label, for example "vacation" **1801**, the system **104** may display sub-labels corresponding to the user selected label. For example, the system **104** display "private" **1811**, "friends" **1812** and "Family" **1813**, as sub-labels corresponding the user selected label "vacation" **1801**.

[0136] FIG. **19** illustrates recommendation of labels for media access request by an application, according to another embodiment.

[0137] In the illustrated embodiment, the application **1900** requests access for storage to access image files stored in the electronic device **100**. The system **104** may suggest labels **1901** corresponding to image files. The system **104** may also allow the user to select image files **1902** to which the user wishes to grant access to the application **102**.

[0138] FIG. **20** illustrates learning recommendation, according to an embodiment.

[0139] For example, based on a storage access request from an application, the system **104** may recommend labels **2000** such as finance **2001**, card **2003** and bank statement **2005**. The system **104** may also recommend some additional labels such as, vacation **2011**, personal **2013**, official **2015**, social **2017** and business **2019**. When a select from additional labels such as "vacation" **2011**, the system **104** may learn user preference based on selected label and recommend labels for future request based on user preference.

[0140] FIG. **21** illustrates a comparison of clipboard content access according to a technique and an embodiment.

[0141] According to the technique, all the information **2110** store in clipboard **2100** may be accessed by any application installed with the electronic device **100**. For example, all clipboard contents are available (visible) to all apps. According to an embodiment of present disclosure, the system **104** may enable a user to allow access to selective information **2120** from the clipboard **2100**, as shown in FIG. **21**. For example, dynamically selected contents are shows to apps.

[0142] In an embodiment, a method (**1500**) for managing access of content for an application in an electronic device is disclosed. The method (**1500**) includes receiving (**1502**), from the application, a content access request to access one or more content types available at the electronic device. The method (**1500**) includes providing (**1504**) one or more labels from a plurality of labels corresponding to each one of the one or more content types requested by the application to a user of the electronic device. The method (**1500**) includes receiving (**1506**) a user selection of at least one label from the one or more labels corresponding to the one or more content types. The method (**1500**) includes providing (**1512**), to the application, an access to the one or more content types requested by the application based on the user selection.

[0143] In an embodiment, the method (**1500**) includes providing (**1508**), one or more sub-labels based on the user selection of the at least one label. The method (**1500**) includes receiving (**1510**) a user selection of at least one sub-label from the one or more sub-labels corresponding to the one or more content types. The method (**1500**) includes providing (**1512**), to the application, the access to the one or more content types requested by the application based on the user selection of the at least one sub-label.

[0144] In an embodiment, the one or more content types comprises at least one of images, videos, text messages, electronic device information, and sensors-related data.

[0145] In an embodiment, the method (**1500**) includes monitoring (**1514**) an actual content type accessed by the application. The method (**1500**) includes determining (**1516**) whether the actual content type is different from the provided access to the one or more content types. The method (**1500**) includes determining (**1518**) a spurious access by the application to alert the user in response to determining that the actual content type accessed by the application is different than the provided access to the one or more content types. The method (**1500**) includes displaying (**1520**) the determined spurious access by the application.

[0146] In an embodiment, the method (**1500**) includes determining (**1522**) a type of the application based on one or more metadata associated with the application. The method (**1500**) includes restricting (**1524**) access to a set of content type for the application based on the type of application.

[0147] In an embodiment, the method (**1500**) includes receiving (**1526**), from the application, a sensor access request to access one or more sensors of the electronic device (**100**). The method (**1500**) includes determining (**1528**) sensor request related data corresponding to the sensor access request by the application. The method (**1500**) includes determining (**1530**) whether the sensor request related data complies with a predefined criteria to access the one or more sensors. The method (**1500**) includes providing (**1532**), to the application, an access to the one or more sensors requested by the application upon determining that the sensor request related data complies with the predefined criteria to access the one or more sensors.

[0148] In an embodiment, the method (**1500**) includes monitoring (**1534**) the access of the one or more sensors by the application. The method (**1500**) includes determining (**1536**) whether the application is in use by the user while accessing the one or more sensors. The method (**1500**) includes restricting (**1538**) the access to the one or more sensors by the application in response to determining that the application is not in use by the user while accessing the one or more sensors.

[0149] In an embodiment, the method (**1500**) includes generating (**1540**) an alert for the user upon determining that the application is not in use by the user while accessing the one or more sensors.

[0150] In an embodiment, the method (**1500**) includes generating (**1542**) pseudo content based on the one or more content type requested by the application. The method (**1500**) includes providing (**1544**) access to the pseudo content to the application.

[0151] In an embodiment, a method (**1600**) for managing access of content for an application in an electronic device (**100**) is disclosed. The method (**1600**) includes receiving (**1602**), from the application, a content access request to access one or more content types available at the electronic device (**100**). The method (**1600**) includes determining (**1604**) request-related data corresponding to the content access request by the application. The method (**1600**) includes determining (**1606**) a plurality of labels associated with each of the one or more content types requested by the application, wherein the plurality of the labels corresponds to a plurality of semantic categories associated with the one or more content types. The method (**1600**) includes providing (**1608**) one or more labels from the plurality of labels corresponding to each one of the one or more content types requested by the application to a user of the electronic device, based on the determined request-related data. The method (**1600**) includes receiving (**1610**) a user selection of at least one label from the one or more labels corresponding to the one or more content types. The method (**1600**) includes providing (**1614**), to the application, an access to the one or more content types requested by the application based on the user selection.

[0152] In an embodiment, the request-related data comprises at least one of a type of the application, a use of requested content by the application, an information associated with access provided to the application based on one or more previous requests, or an information associated with access provided to one or more applications similar to the application.

[0153] In an embodiment, the plurality of semantic categories associated with the one or more content types is based on a demographic profile of the user.

[0154] In an embodiment, the demographic profile of the user is based on at least one of a location, an age, a sex, race, and user-interests.

[0155] In an embodiment, the method (**1600**) includes providing (**1611**), one or more sub-labels based on the user selection of the at least one label. The method (**1600**) includes receiving (**1612**) a user selection of at least one sub-label from the one or more sub-labels corresponding to the one or more content types. The method (**1600**) includes providing (**1614**), to the application, the access to the one or more content types requested by the application based on the user selection of the at least one sub-label.

[0156] In an embodiment, the one or more content types comprises at least one of images, videos, text messages, electronic device information, and sensors-related data.

[0157] In an embodiment, the method (**1600**) includes learning (**1616**) user preferences based on user selections of the at least one label for one or more access requests for one or more content types by the application. The method (**1600**) includes updating (**1618**) the one or more semantic categories for the one or more content types based on the user preferences.

[0158] In an embodiment, the method (**1600**) includes determining (**1620**) a rank associated with each of the plurality of labels associated with each of the one or more content types requested by the application based on the learned user preferences. The method (**1600**) includes providing (**1622**) the one or more labels associated corresponding to each one of the one or more content types based on the determined rank corresponding to each label.

[0159] In an embodiment, the method (**1600**) includes monitoring (**1624**) an actual content type accessed by the application. The method (**1600**) includes determining (**1626**) whether the actual content type is different from the provided access to the one or more content types. The method (**1600**) includes determining (**1628**) a spurious access by the application to alert the user in response to determining that the actual content type accessed by the application is different than the provided access to the one or more content types. The method (**1600**) includes displaying (**1630**) the determined spurious access by the application.

[0160] In an embodiment, the method (**1600**) includes determining (**1632**) a type of the application based on one or more metadata associated with the application. The method (**1600**) includes restricting (**1634**) access to a set of content type for the application based on the type of application.

[0161] In an embodiment, the method (**1600**) includes receiving (**1636**), from the application, a sensor access request to access one or more sensors of the electronic device. The method (**1600**) includes determining (**1638**) sensor request related data corresponding to the sensor access request by the application. The method (**1600**) includes determining (**1640**) whether the sensor request related data complies with a predefined criteria to access the one or more sensors. The method (**1600**) includes providing (**1642**), to the application, an access to the one or more sensors requested by the application upon determining that the sensor request related data complies with the predefined criteria to access the one or more sensors.

[0162] In an embodiment, the method (**1600**) includes monitoring (**1644**) the access of the one or more sensors by the application. The method (**1600**) includes determining (**1646**) whether the application is in use by the user while accessing the one or more sensors. The method (**1600**) includes restricting (**1648**) the access to the one or more sensors by the application in response to determining that the application is not in use by the user while accessing the one or more sensors.

[0163] In an embodiment, the method (**1600**) includes generating (**1650**) an alert for the user upon determining that the application is not in use by the user while accessing the one or more sensors.

[0164] In an embodiment, the method (**1600**) includes generating (**1652**) pseudo content based on the one or more content type requested by the application. The method (**1600**) includes providing (**1654**) access to the pseudo content to the application.

[0165] In an embodiment, an electronic device (**100**) for managing access to an application is disclosed. The electronic device (**100**) includes a memory (**210**) and at least one processor (**202**) communicably coupled to the memory. The at least one processor (**202**) is configured to receive, from the application, a content access request to access one or more content types available at the electronic device (**100**). The at least one processor (**202**) is configured to provide one or more labels from a plurality of labels corresponding to each one of the one or more content types requested by the application to a user of the electronic device (**100**). The at least one processor (**202**) is configured to receive a user selection of at least one label from the one or more labels corresponding to the one or more content types. The at least one processor (**202**) is configured to provide, to the application, an access to the one or more content types requested by the application based on the user selection.

[0166] In an embodiment, the at least one processor (**202**) is configured to provide, one or more sub-labels based on the user selection of the at least one label. The at least one processor (**202**) is configured to receive a user selection of at least one sub-label from the one or more sub-labels corresponding to the one or more content types. The at least one processor (**202**) is configured to provide, to the application, the access to the one or more content types requested by the application based on the user selection of the at least one sub-label.

[0167] In an embodiment, the one or more content types comprises at least one of images, videos, text messages, electronic device information, and sensors-related data.

[0168] In an embodiment, the at least one processor (**202**) is configured to monitor an actual content type accessed by the application. The at least one processor (**202**) is configured to determine whether the actual content type is different from the provided access to the one or more content types. The at least one processor (**202**) is configured to determine a spurious access by the application to alert the user in response to determining that the actual content type accessed by the application is different than the provided access to the one or more content types. The at least one processor (**202**) is configured to display the determined spurious access by the application.

[0169] In an embodiment, the at least one processor (**202**) is configured to determine a type of the application based on one or more metadata associated with the application. The at least one processor (**202**) is configured to restrict access to a set of content type for the application based on the type of application.

[0170] In an embodiment, the at least one processor (**202**) is configured to receive, from the application, a sensor access request to access one or more sensors of the electronic device (**100**). The at least one processor (**202**) is configured to determine sensor request related data corresponding to the sensor access request by the application. The at least one processor (**202**) is configured to determine whether the sensor request related data complies with a predefined criteria to access the one or more sensors. The at least one processor (**202**) is configured to provide, to the application, an access to the one or more sensors requested by the application upon determining that the sensor request related data complies with the predefined criteria to access the one or more sensors.

[0171] In an embodiment, the at least one processor (**202**) is configured to monitor the access of the one or more sensors by the application. The at least one processor (**202**) is configured to determine whether the application is in use by the user while accessing the one or more sensors. The at least one processor (**202**) is configured to restrict the access to the one or more sensors by the application in response to determining that the application is not in use by the user while accessing the one or more sensors.

[0172] In an embodiment, the at least one processor (**202**) is configured to generate an alert for the user upon determining that the application is not in use by the user while accessing the one or more sensors.

[0173] In an embodiment, the at least one processor (**202**) is configured to generate pseudo content based on the one or more content type requested by the application. The at least one processor (**202**) is configured to provide access to the pseudo content to the application.

[0174] In an embodiment, an electronic device (**100**) for managing access of content for an application in the electronic device (**100**) is disclosed. The electronic device (**100**) includes a memory (**210**) and at least one processor (**202**) communicably coupled to the memory (**210**). The at least one processor (**202**) is configured to receive, from the application, a content access request to access one or more content types available at the electronic device (**100**). The at least one processor (**202**) is configured to determine request-related data corresponding to the content access request. The at least one processor (**202**) is configured to determine a plurality of labels associated with each of the one or more content types requested by the application, wherein the plurality of the labels corresponds to a plurality of semantic categories associated with the one or more content types. The at least one processor (**202**) is configured to provide one or more labels from the plurality of labels corresponding to each one of the one or more content types requested by the application to a user of the electronic device, based on the determined request-related data. The at least one processor (**202**) is configured to receive a user selection of at least one label from the one or more labels corresponding to the one or more content types. The at least one processor (**202**) is configured to provide, to the application, an access to the one or more content types requested by the application based on the user selection.

[0175] In an embodiment, the request-related data comprises at least one of a type of the application, a use of requested content by the application, an information associated with access provided to the application based on one or more previous requests, or an information associated with access provided to one or more applications similar to the application.

[0176] In an embodiment, the plurality of semantic categories associated with the one or more content types is based on a demographic profile of the user.

[0177] In an embodiment, the demographic profile of the user is based on at least one of a location, an age, a sex, race, and user-interests.

[0178] In an embodiment, the at least one processor (**202**) is configured to provide, one or more sub-labels based on the user selection of the at least one label. The at least one processor (**202**) is configured to receive a user selection of at least one sub-label from the one or more sub-labels corresponding to the one or more content types. The at least one processor (**202**) is configured to provide, to the application, the access to the one or more content types requested by the application based on the user selection of the at least one sub-label.

[0179] In an embodiment, the one or more content types comprises at least one of images, videos, text messages, electronic device information, and sensors-related data.

[0180] In an embodiment, the at least one processor (**202**) is configured to learn user preferences based on user selections of the at least one label for one or more access requests for one or more content types by the application. The at least one processor (**202**) is configured to update the one or more semantic categories for the one or more content types based on the user preferences.

[0181] In an embodiment, the at least one processor (**202**) is configured to determine a rank associated with each of the plurality of labels associated with each of the one or more content types requested by the application based on the learned user preferences. The at least one processor (**202**) is configured to provide the one or more labels associated corresponding to each one of the one or more content types based on the determined rank corresponding to each label.

[0182] In an embodiment, the at least one processor (**202**) is configured to monitor an actual content type accessed by the application. The at least one processor (**202**) is configured to determine whether the actual content type is different from the provided access to the one or more content types. The at least one processor (**202**) is configured to determine a spurious access by the application to alert the user in response to determining that the actual content type accessed by the application is different than the provided access to the one or more content types. The at least one processor (**202**) is configured to display the determined spurious access by the application.

[0183] In an embodiment, the at least one processor (**202**) is configured to determine a type of the application based on one or more metadata associated with the application. The at least one processor (**202**) is configured to restrict access to a set of content type for the application based on the type of application.

[0184] In an embodiment, the at least one processor (**202**) is configured to receive, from the application, a sensor access request to access one or more sensors of the electronic device. The at least one processor (**202**) is configured to determine sensor request related data corresponding to the sensor access request by the application. The at least one processor (**202**) is configured to determine whether the sensor request related data complies with a predefined criteria to access the one or more sensors. The at least one processor (**202**) is configured to provide, to the application, an access to the one or more sensors requested by the application upon determining that the sensor request related data complies with the predefined criteria to access the one or more sensors.

[0185] In an embodiment, the at least one processor (**202**) is configured to monitor the access of the one or more sensors by the application. The at least one processor (**202**) is configured to determine whether the application is in use by the user while accessing the one or more sensors. The at least one processor (**202**) is configured to restrict the access to the one or more sensors by the application in response to determining that the application is not in use by the user while accessing the one or more sensors.

[0186] In an embodiment, the at least one processor (**202**) is configured to generate an alert for the user upon determining that the application is not in use by the user while accessing the one or more sensors.

[0187] In an embodiment, the at least one processor (**202**) is configured to generate pseudo content based on the one or more content type requested by the application. The at least one processor (**202**) is configured to provide access to the pseudo content to the application.

[0188] The present disclosure provides an application permission model based on semantic content classification. The present disclosure may facilitate selecting content-based permissions for an application. The present disclosure may also facilitate permission recommendations for the user based on the application, requested access and user demographic profile. The disclosure may increase privacy while using an application. By logging access information from the application, the disclosure may also reduce data synchronization by the application to the server. The disclosure may reduce the risk of data misuse while providing data cleaning functionalities that may facilitate removal of data from the electronic device.

[0189] While language has been used to describe the present subject matter, any limitations arising on account thereto, are not intended. As would be apparent to a person in the art, various modifications may be made to the method within the scope of the disclosure. The drawings and the foregoing description give examples of embodiments. Those skilled in the art will appreciate that one or more of the described elements may be combined into a single functional element. Certain elements may be split into multiple functional elements. Elements from one embodiment may be added to another embodiment.

What is claimed is:

1. A method for managing access to content for an application of an electronic device, the method comprising:

receiving, from the application, a content access request for granting access to first data of one or more content types available via the electronic device;

displaying one or more labels, from among a plurality of labels, corresponding to the one or more content types;

receiving a first user selection of at least one label, from among the one or more labels, corresponding to the one or more content types; and

granting, to the application, access to the first data of the one or more content types based on the first user selection.

2. The method as claimed in claim **1**, further comprising:

displaying one or more sub-labels based on the first user selection;

receiving a second user selection of at least one sub-label, from among the one or more sub-labels, corresponding to at least one of the one or more content types; and

granting, to the application, access to second data of the at least one of the one or more content types based on the second user selection.

3. The method as claimed in claim **1**, wherein the one or more content types comprise at least one of images, videos, text messages, electronic device information, or sensor data.

4. The method as claimed in claim **1**, further comprising:

monitoring a content type of second data accessed by the application;

determining whether the content type deviates from the access granted;

detecting a spurious access by the application based on the content type deviating from the access granted; and

displaying a notification indicating the spurious access.

5. The method as claimed in claim **1**, further comprising:

determining an application type based on metadata associated with the application; and

restricting the access to second data from among a set of one or more content types based on the application type.

6. The method as claimed in claim **1**, further comprising:

receiving, from the application, a sensor access request to access one or more sensors of the electronic device;

determining second data corresponding to the sensor access request by the application;

determining whether the second data complies with a predefined criteria for accessing the one or more sensors; and

providing, to the application, access to the one or more sensors based on determining that the second data complies with the predefined criteria.

7. The method as claimed in claim **6**, further comprising:

monitoring the access to the one or more sensors by the application;

determining whether the application is in use by a user of the electronic device while the one or more sensors are being accessed by the application; and

based on determining the application is not in use, restricting the access to the one or more sensors by the application.

8. The method as claimed in claim **7**, further comprising:

displaying a notification based on the application not being in use by the user while the one or more sensors are being accessed by the application.

9. The method as claimed in claim **1**, further comprising:

generating pseudo content based on the one or more content types; and

providing the application access to the pseudo content.

10. An electronic device for managing access to content for an application, the electronic device comprising:

memory storing instructions; and

at least one processor;

wherein the instructions, when executed by the at least one processor, cause the electronic device to:

receive, from the application, a content access request for granting access to first data of one or more content types available via the electronic device;

display one or more labels, from among a plurality of labels, corresponding to the one or more content types;

receive a first user selection of at least one label, from among the one or more labels, corresponding to the one or more content types; and

granting, to the application, access to the first data of the one or more content types based on the first user selection.

11. The electronic device as claimed in claim **10**, wherein the instructions, when executed by the at least one processor, cause the electronic device to:

display one or more sub-labels based on the first user selection;

receive a second user selection of at least one sub-label, from among the one or more sub-labels, corresponding to at least one of the one or more content types; and

grant, to the application, access to second data of the at least one of the one or more content types based on the second user selection.

12. The electronic device as claimed in claim **10**, wherein the one or more content types comprise at least one of images, videos, text messages, electronic device information, or sensor data.

13. The electronic device as claimed in claim **10**, wherein the instructions, when executed by the at least one processor, cause the electronic device to:

monitor a content type of second data accessed by the application;

determine whether the content type deviates from the access granted;

detecting a spurious access by the application based on the content type deviating from the access granted; and

display a notification indicating the spurious access.

14. The electronic device as claimed in claim **10**, wherein the instructions, when executed by the at least one processor, cause the electronic device to:

determine an application type based on metadata associated with the application; and

restrict the access to second data from among a set of one or more content types based on the application type.

15. The electronic device as claimed in claim **10**, wherein the instructions, when executed by the at least one processor, cause the electronic device to:

receive, from the application, a sensor access request to access one or more sensors of the electronic device;

determine second data corresponding to the sensor access request by the application;

determine whether the second data complies with a predefined criteria for accessing the one or more sensors; and

provide, to the application, access to the one or more sensors based on determining that the second data complies with the predefined criteria.

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